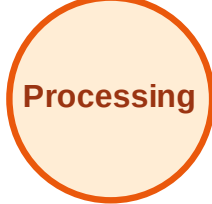


BFS Traversal

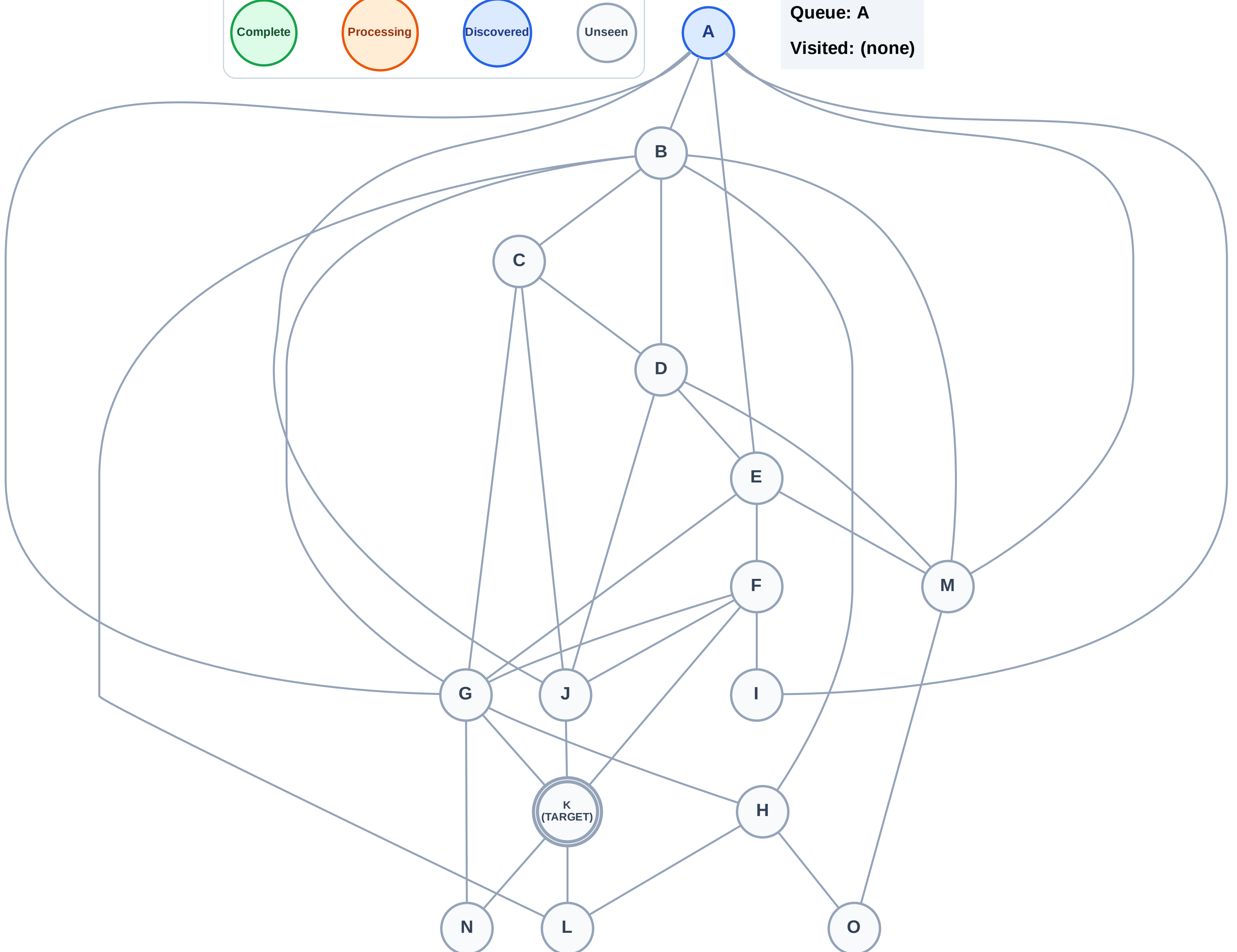
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

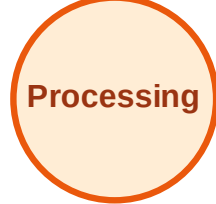
Visited: (none)



BFS Traversal

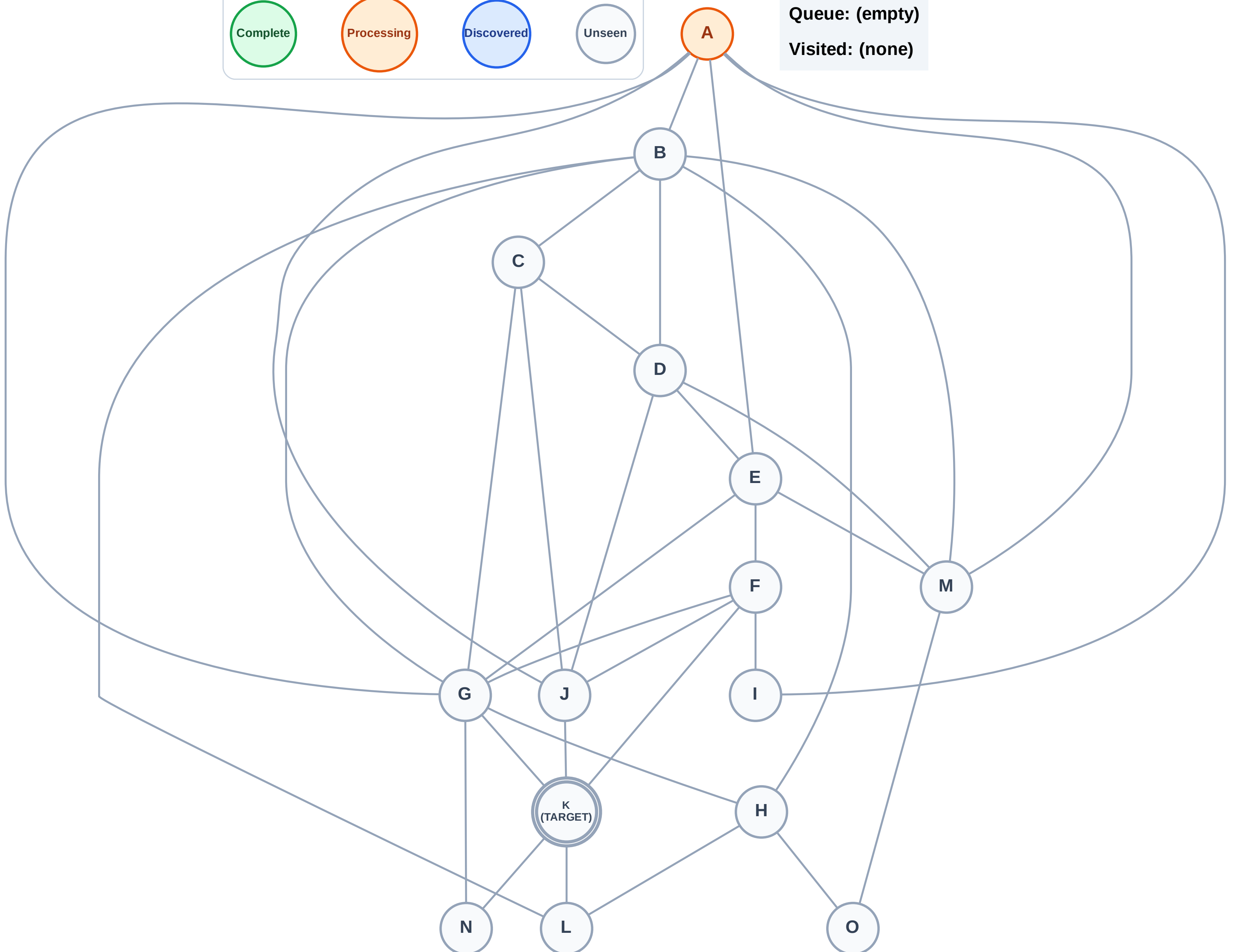
Step 2: Process node A

Legend



Queue: (empty)

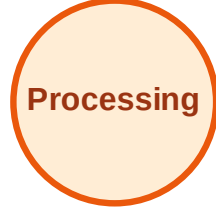
Visited: (none)



BFS Traversal

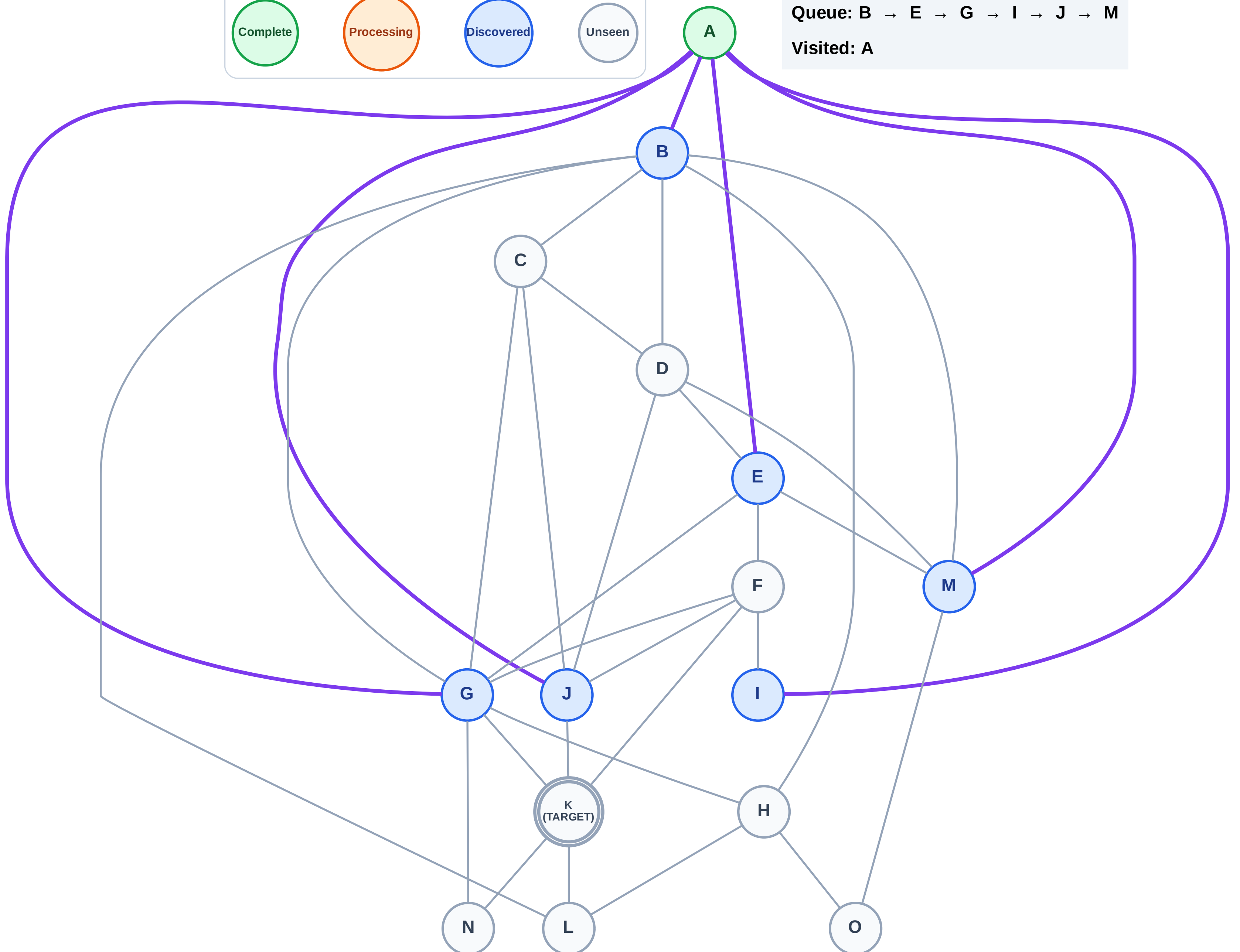
Step 3: Discover B, E, G, I, J, M from A

Legend



Queue: B → E → G → I → J → M

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

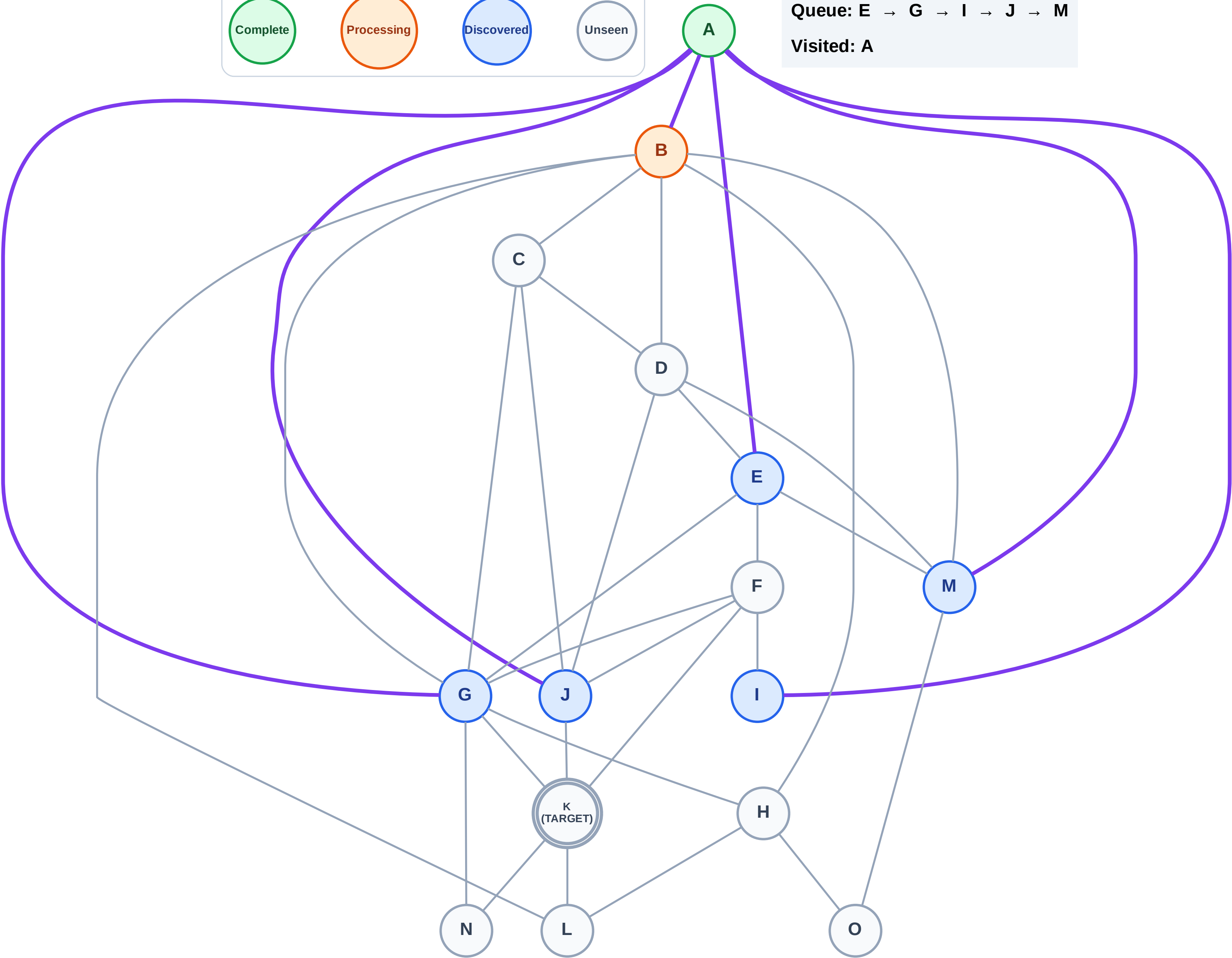
Processing

Discovered

Unseen

Queue: E → G → I → J → M

Visited: A



BFS Traversal

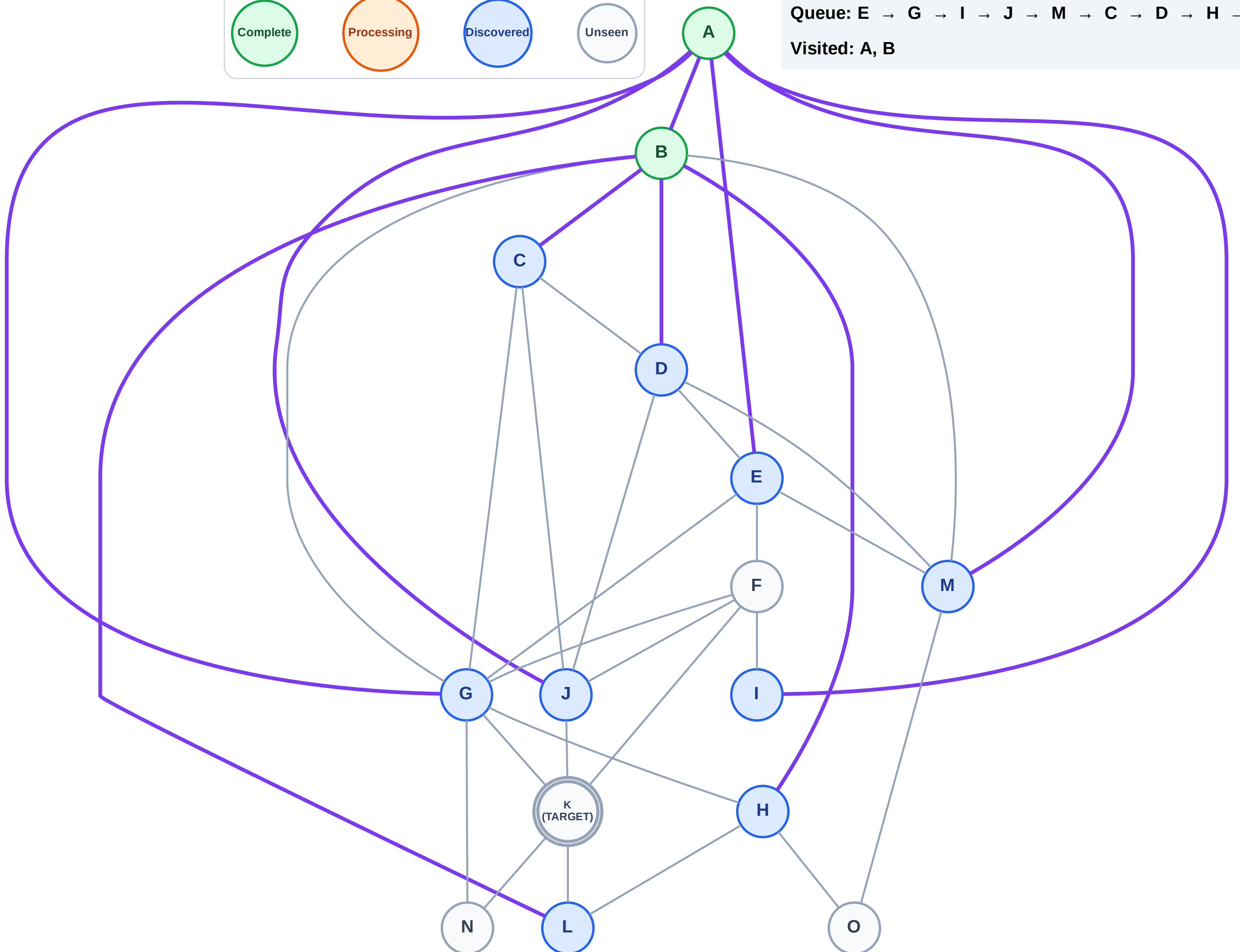
Step 5: Discover C, D, H, L from B

Legend



Queue: E → G → I → J → M → C → D → H → L

Visited: A, B



BFS Traversal

Step 6: Process node E

Legend

Complete

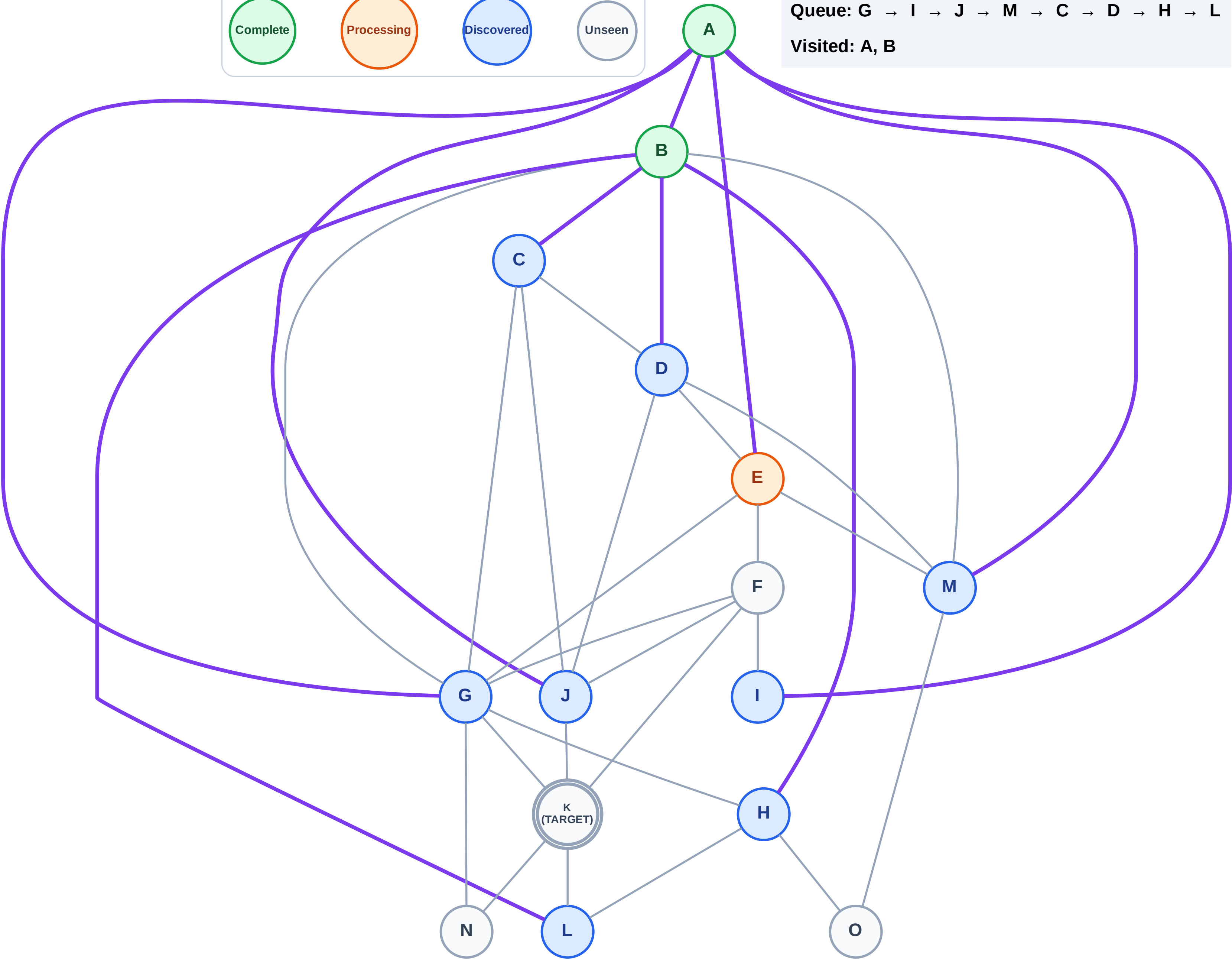
Processing

Discovered

Unseen

Queue: G → I → J → M → C → D → H → L

Visited: A, B



BFS Traversal

Step 7: Discover F from E

Legend



Complete



Processing



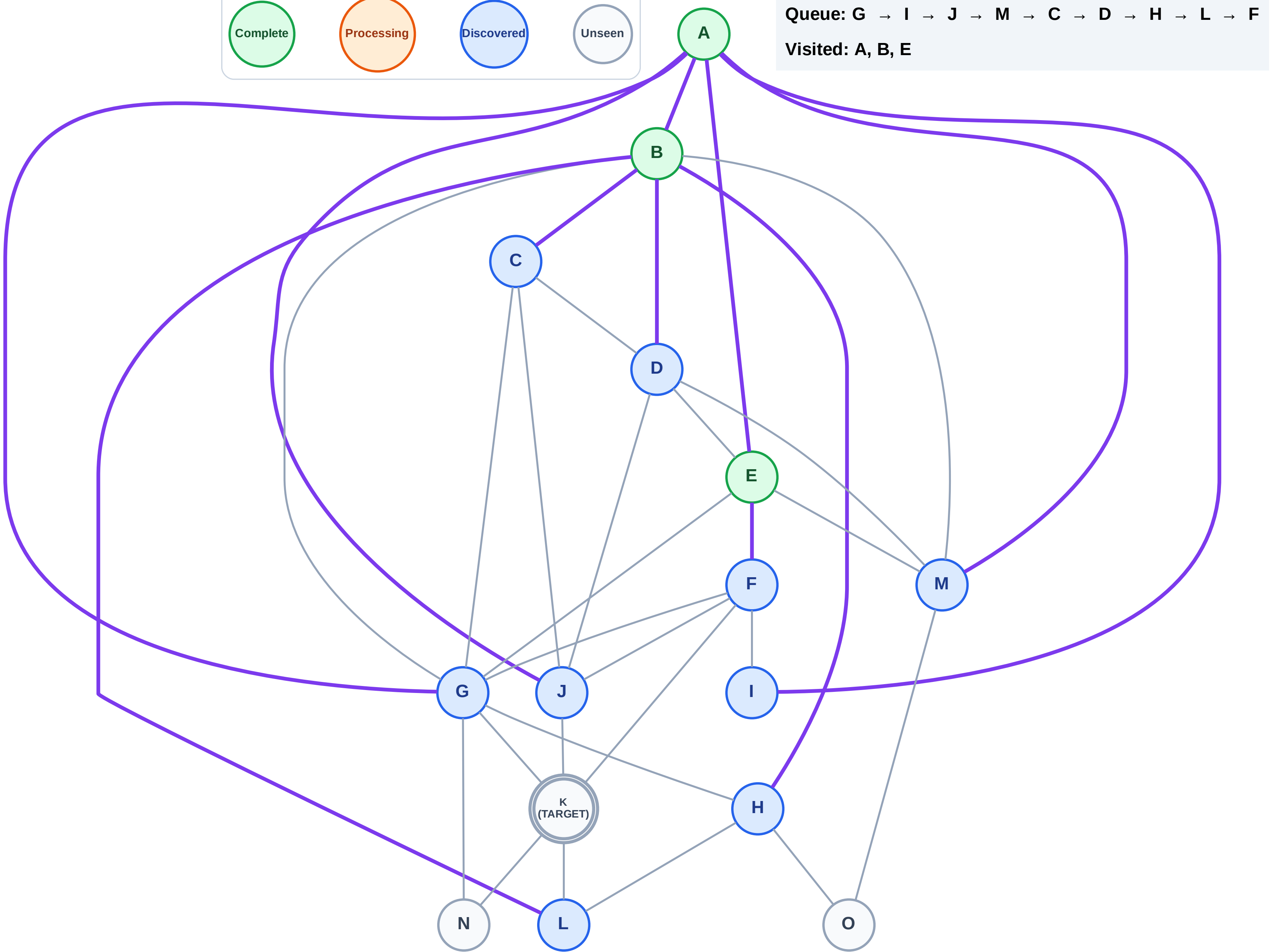
Discovered



Unseen

Queue: G → I → J → M → C → D → H → L → F

Visited: A, B, E



BFS Traversal

Step 8: Process node G

Legend

Complete

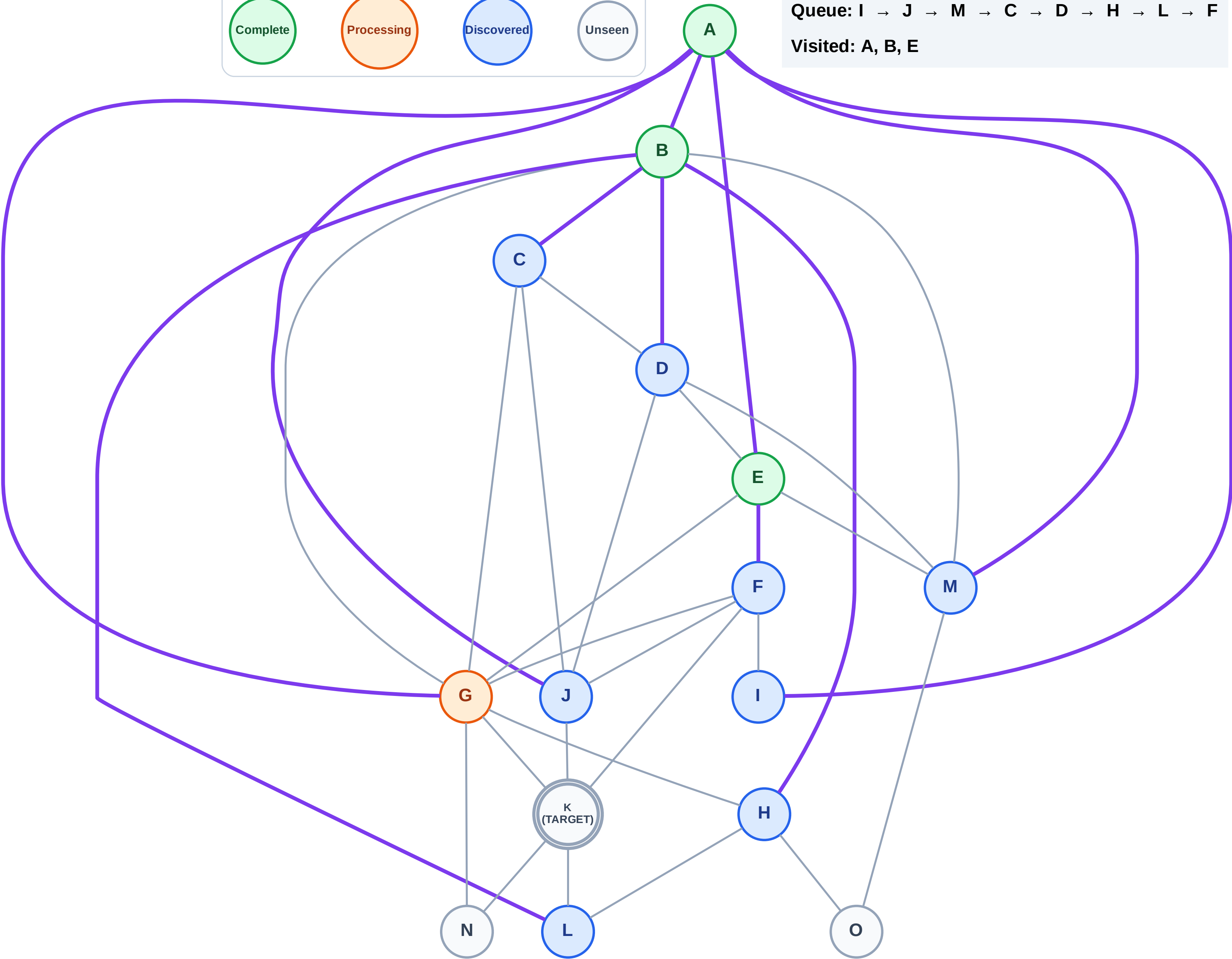
Processing

Discovered

Unseen

Queue: I → J → M → C → D → H → L → F

Visited: A, B, E



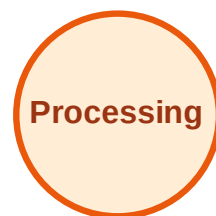
BFS Traversal

Step 9: Discover K, N from G

Legend



Complete



Processing



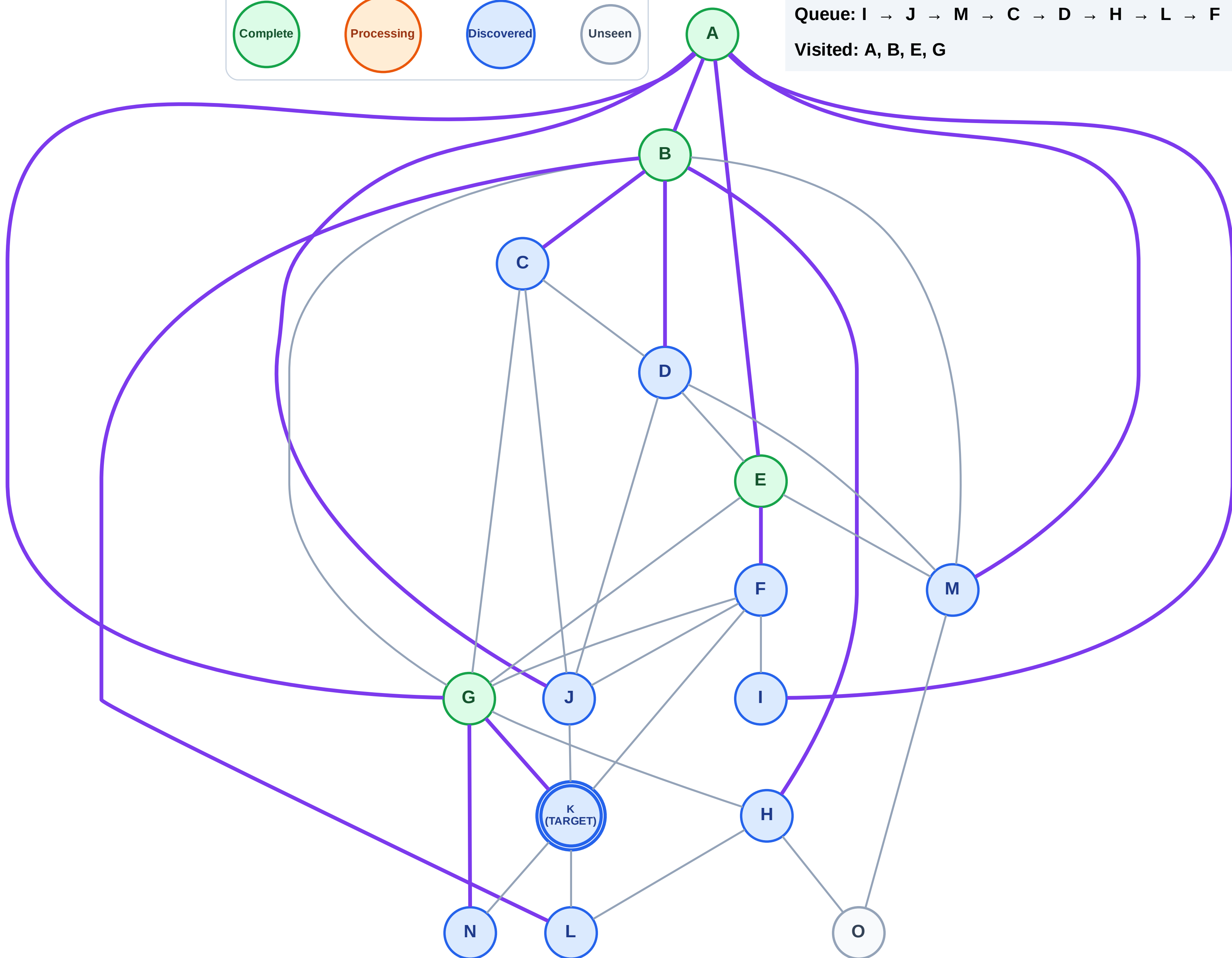
Discovered



Unseen

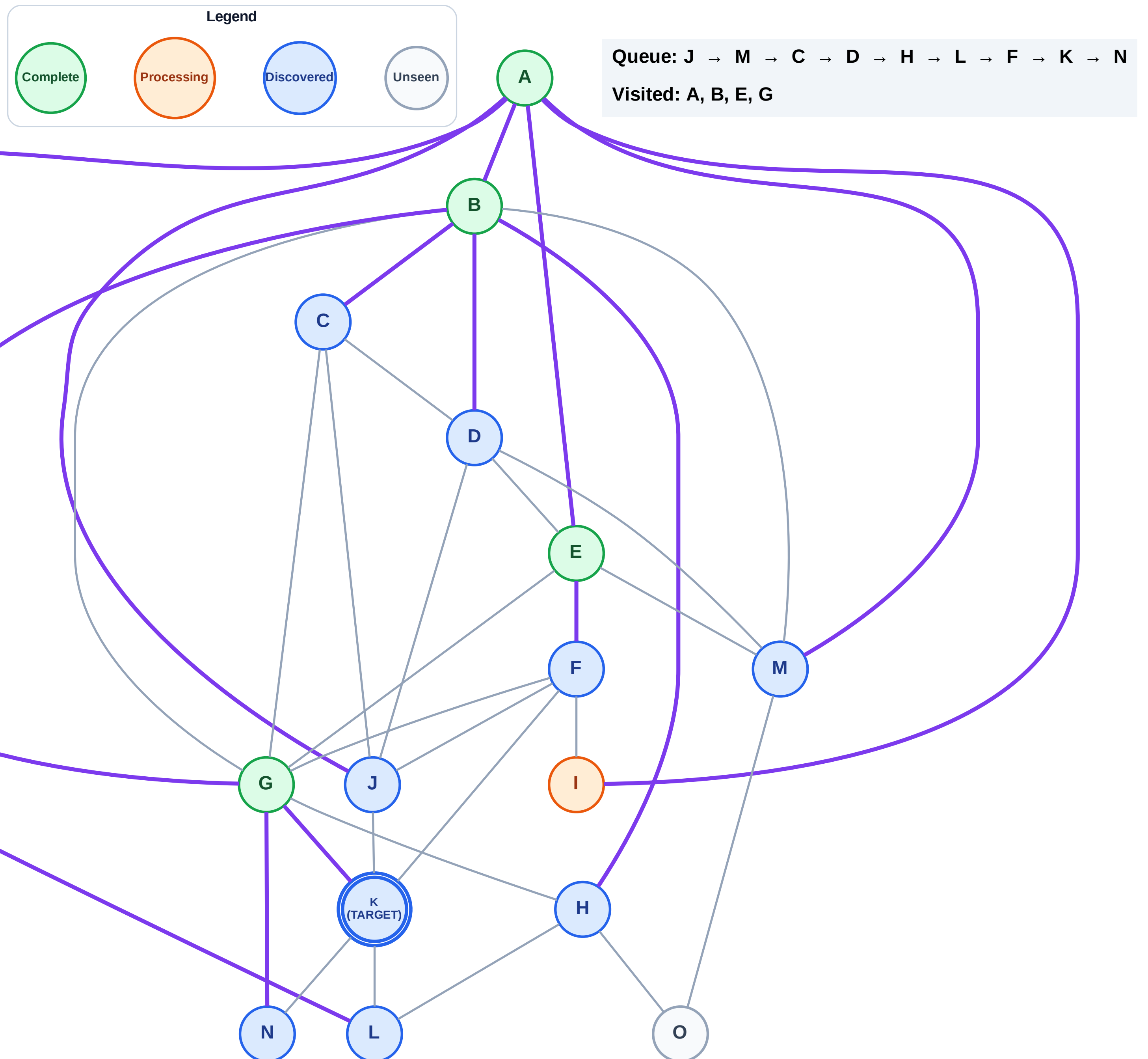
Queue: I → J → M → C → D → H → L → F → K → N

Visited: A, B, E, G



Step 10: Process node I

Step 10: Process node I



BFS Traversal

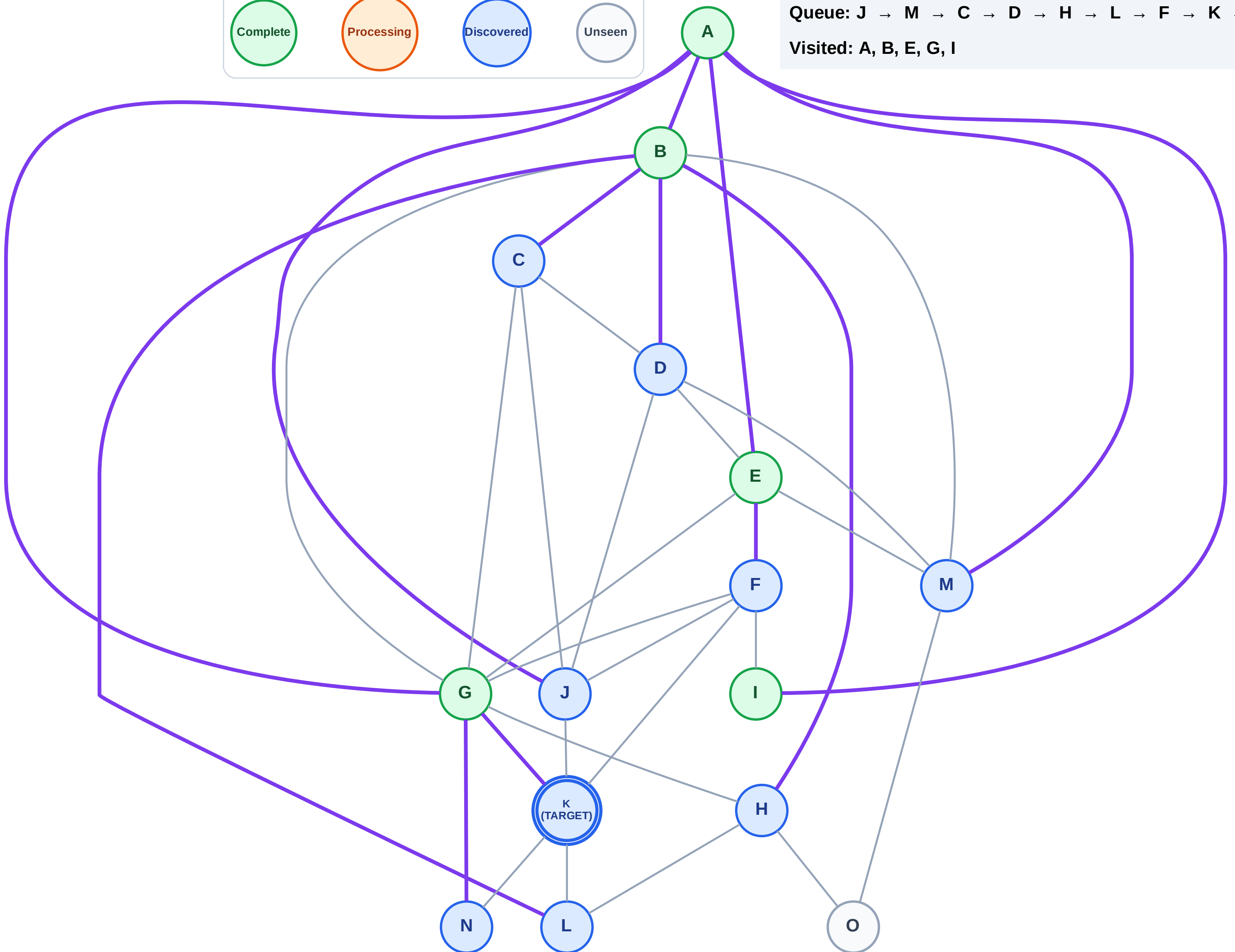
Step 11: No new neighbors from I

Legend



Queue: J → M → C → D → H → L → F → K → N

Visited: A, B, E, G, I



BFS Traversal

Step 12: Process node J

Legend

Complete

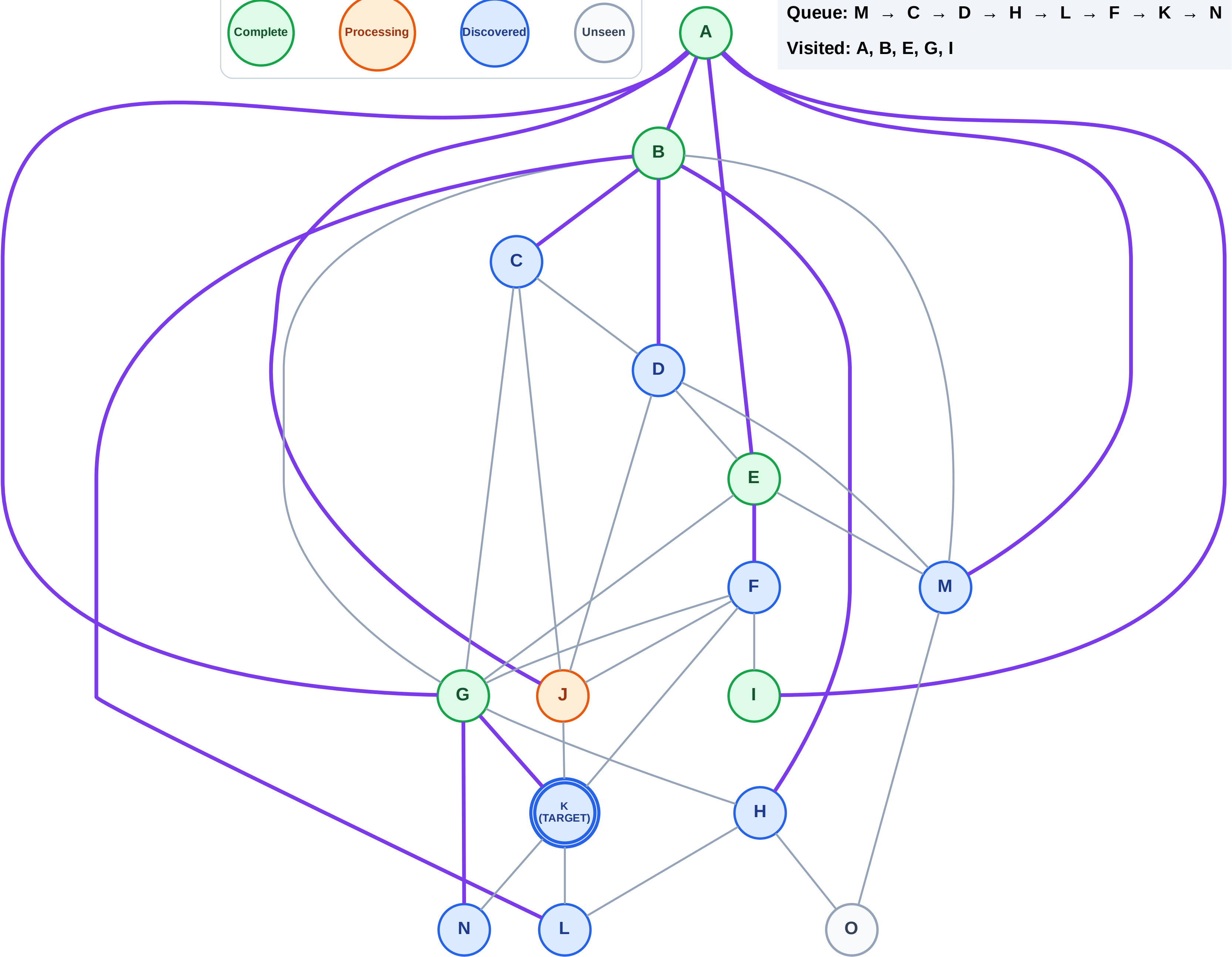
Processing

Discovered

Unseen

Queue: M → C → D → H → L → F → K → N

Visited: A, B, E, G, I



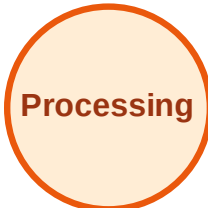
BFS Traversal

Step 13: No new neighbors from J

Legend



Complete



Processing



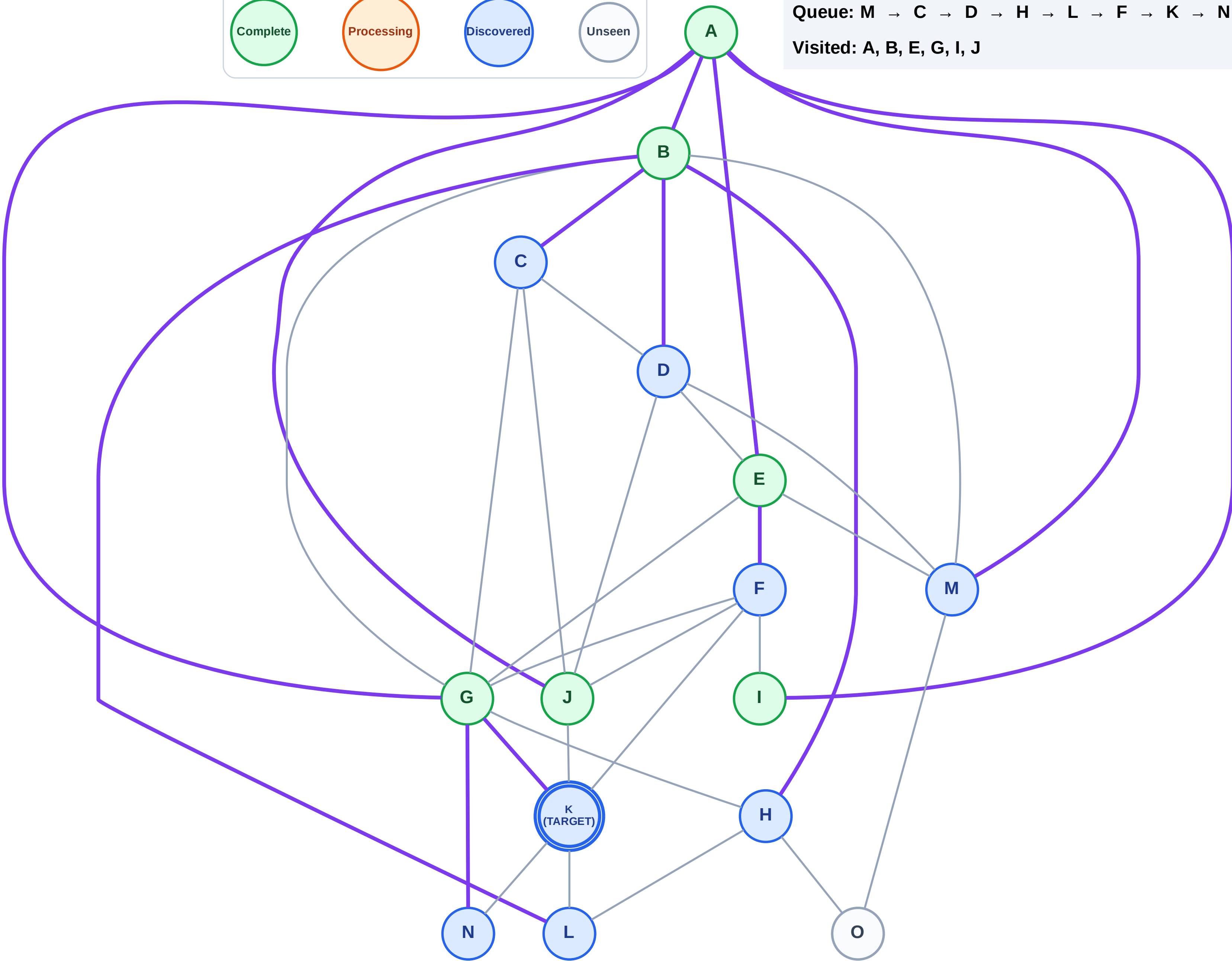
Discovered



Unseen

Queue: M → C → D → H → L → F → K → N

Visited: A, B, E, G, I, J



BFS Traversal

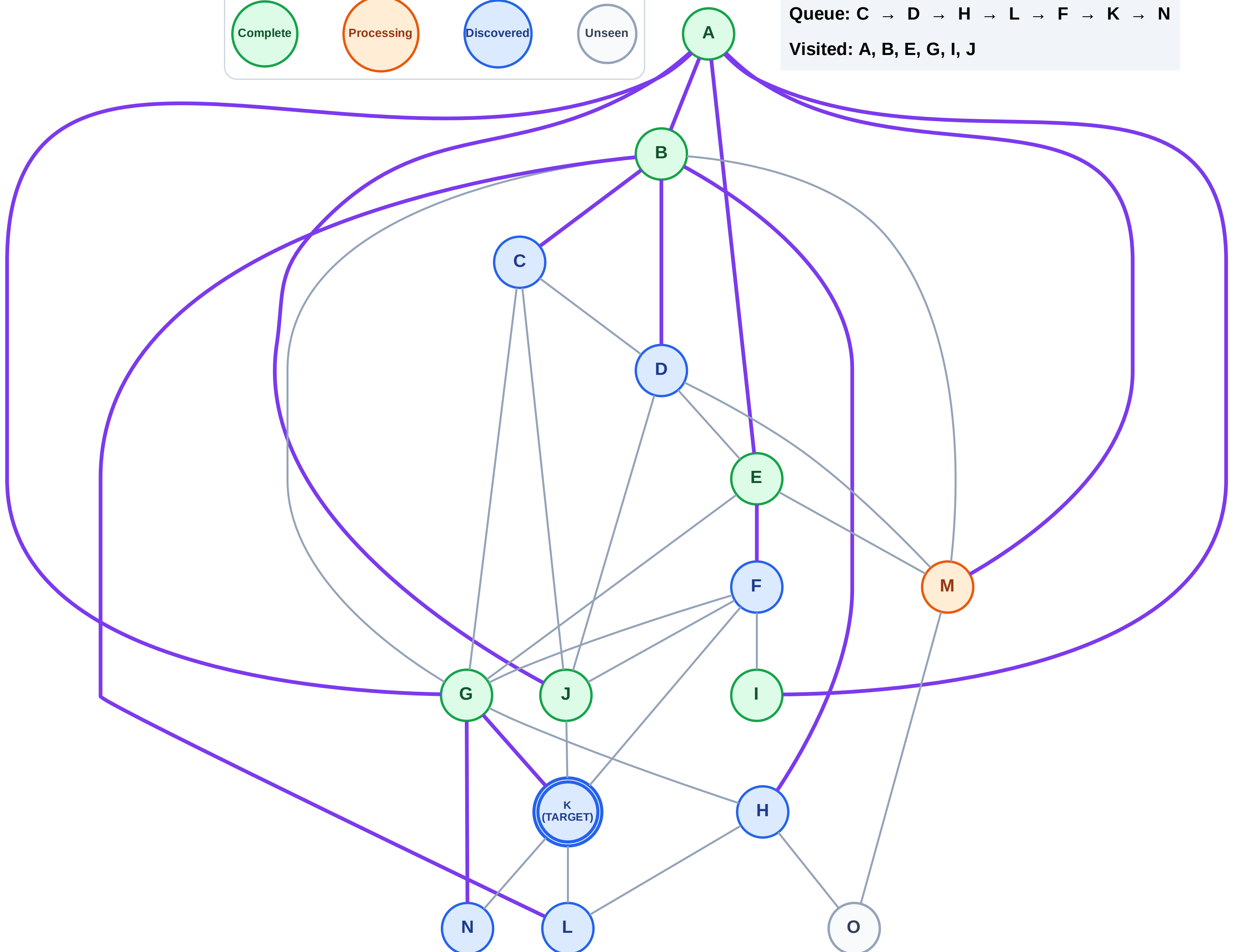
Step 14: Process node M

Legend



Queue: C → D → H → L → F → K → N

Visited: A, B, E, G, I, J



BFS Traversal

Step 15: Discover O from M

Legend



Complete



Processing



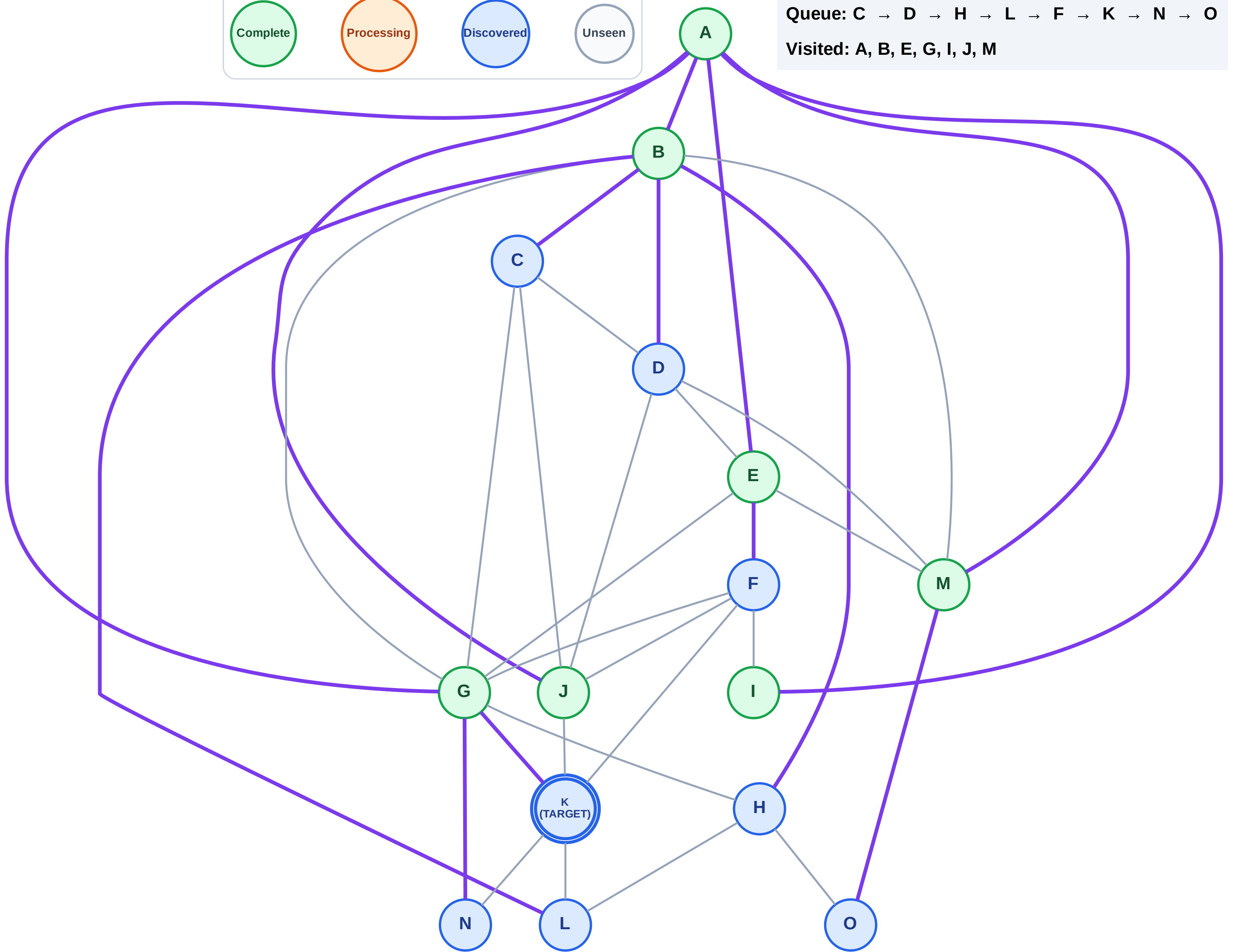
Discovered



Unseen

Queue: C → D → H → L → F → K → N → O

Visited: A, B, E, G, I, J, M



BFS Traversal

Step 16: Process node C

Legend



Complete



Processing



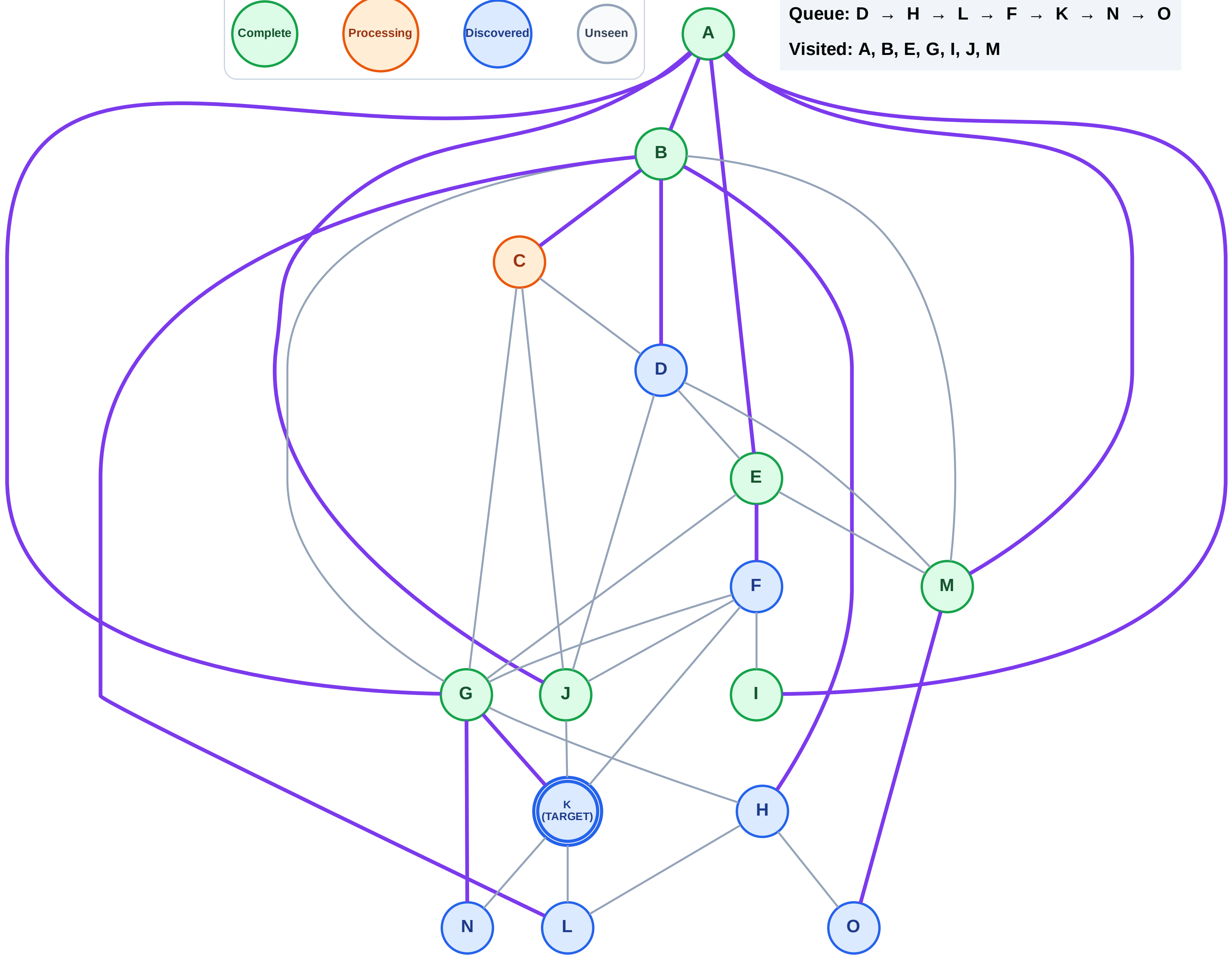
Discovered



Unseen

Queue: D → H → L → F → K → N → O

Visited: A, B, E, G, I, J, M



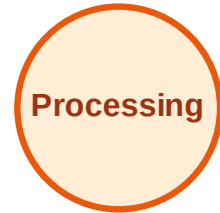
BFS Traversal

Step 17: No new neighbors from C

Legend



Complete



Processing



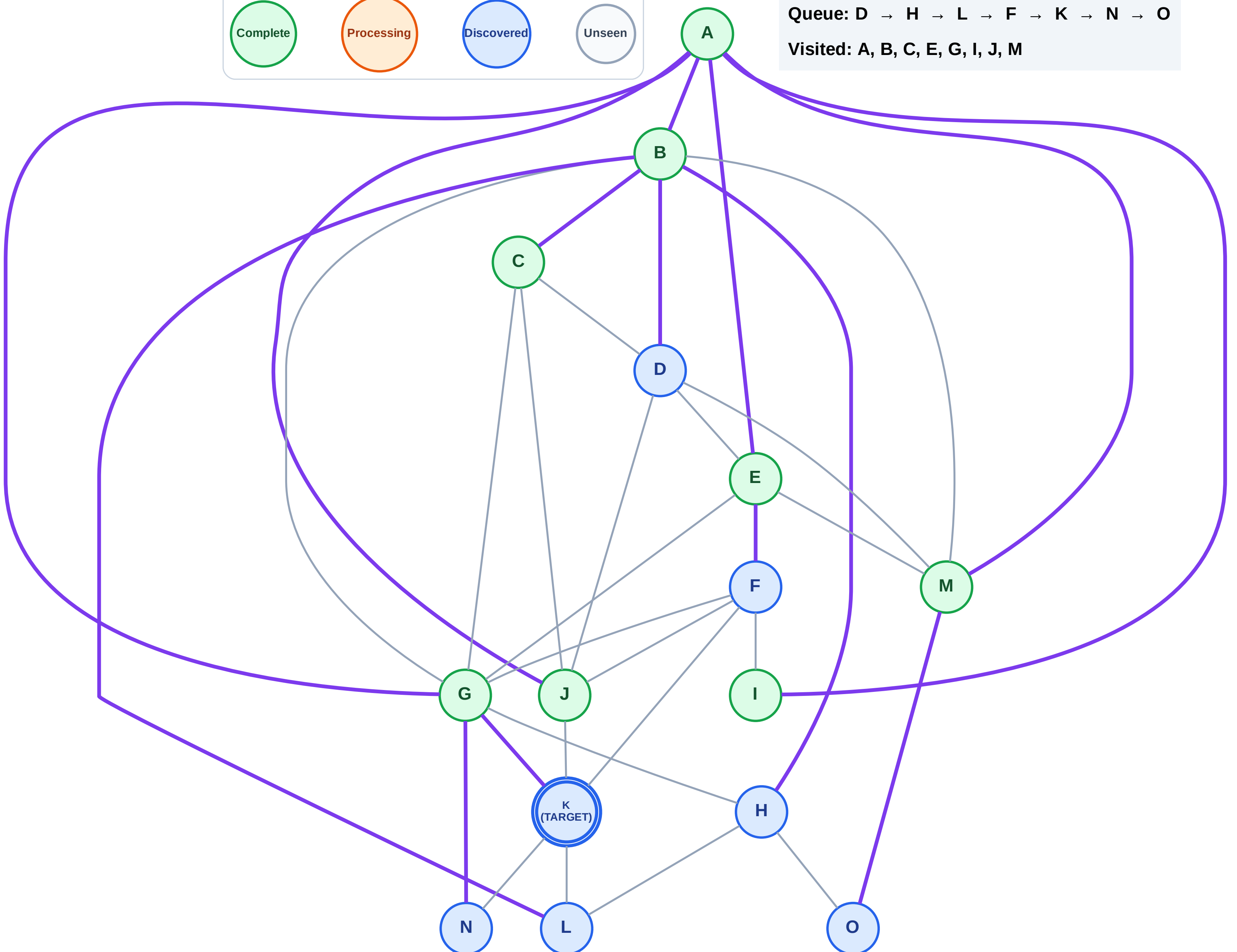
Discovered



Unseen

Queue: D → H → L → F → K → N → O

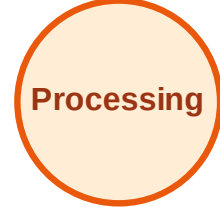
Visited: A, B, C, E, G, I, J, M



BFS Traversal

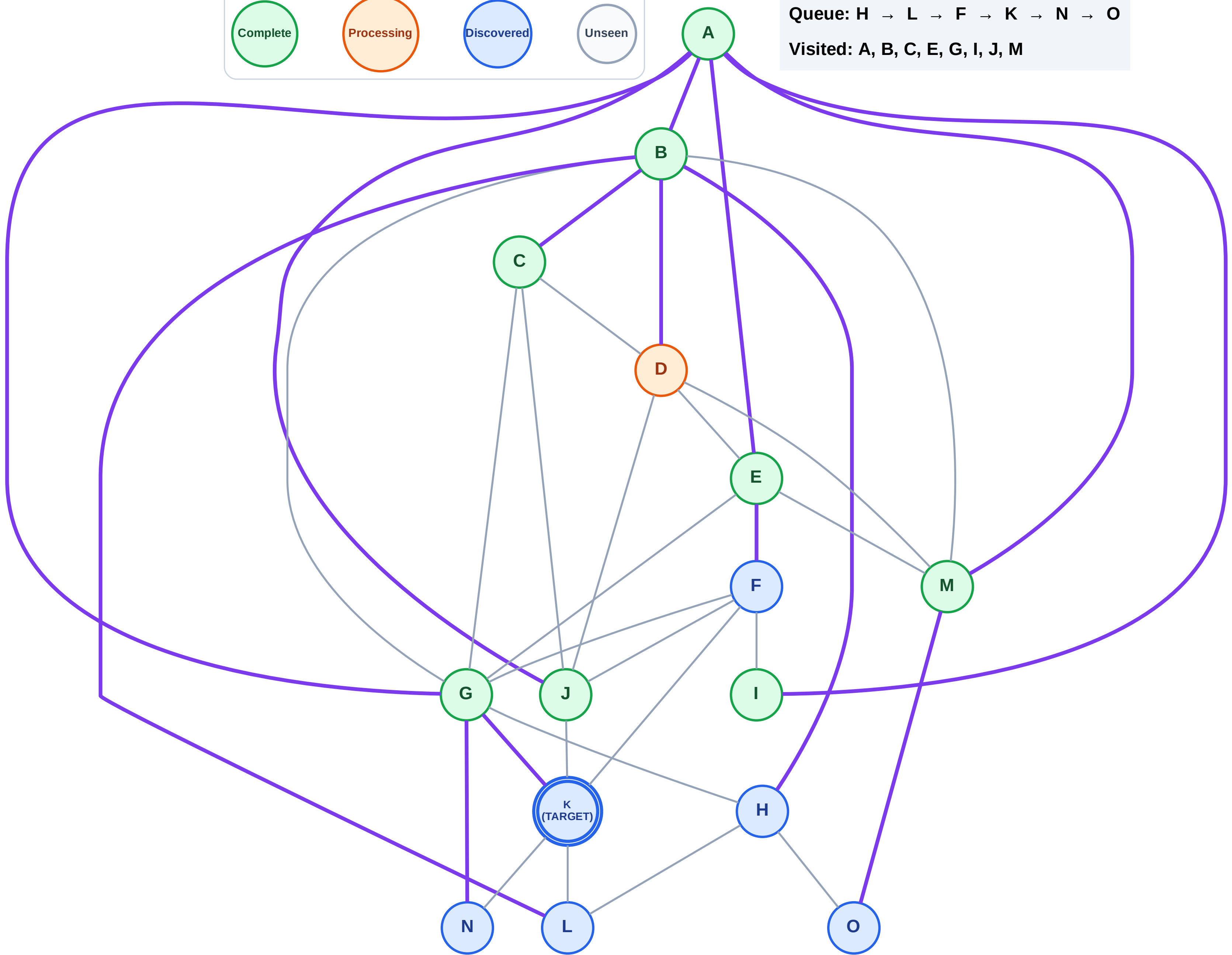
Step 18: Process node D

Legend



Queue: H → L → F → K → N → O

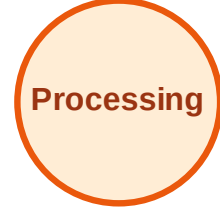
Visited: A, B, C, E, G, I, J, M



BFS Traversal

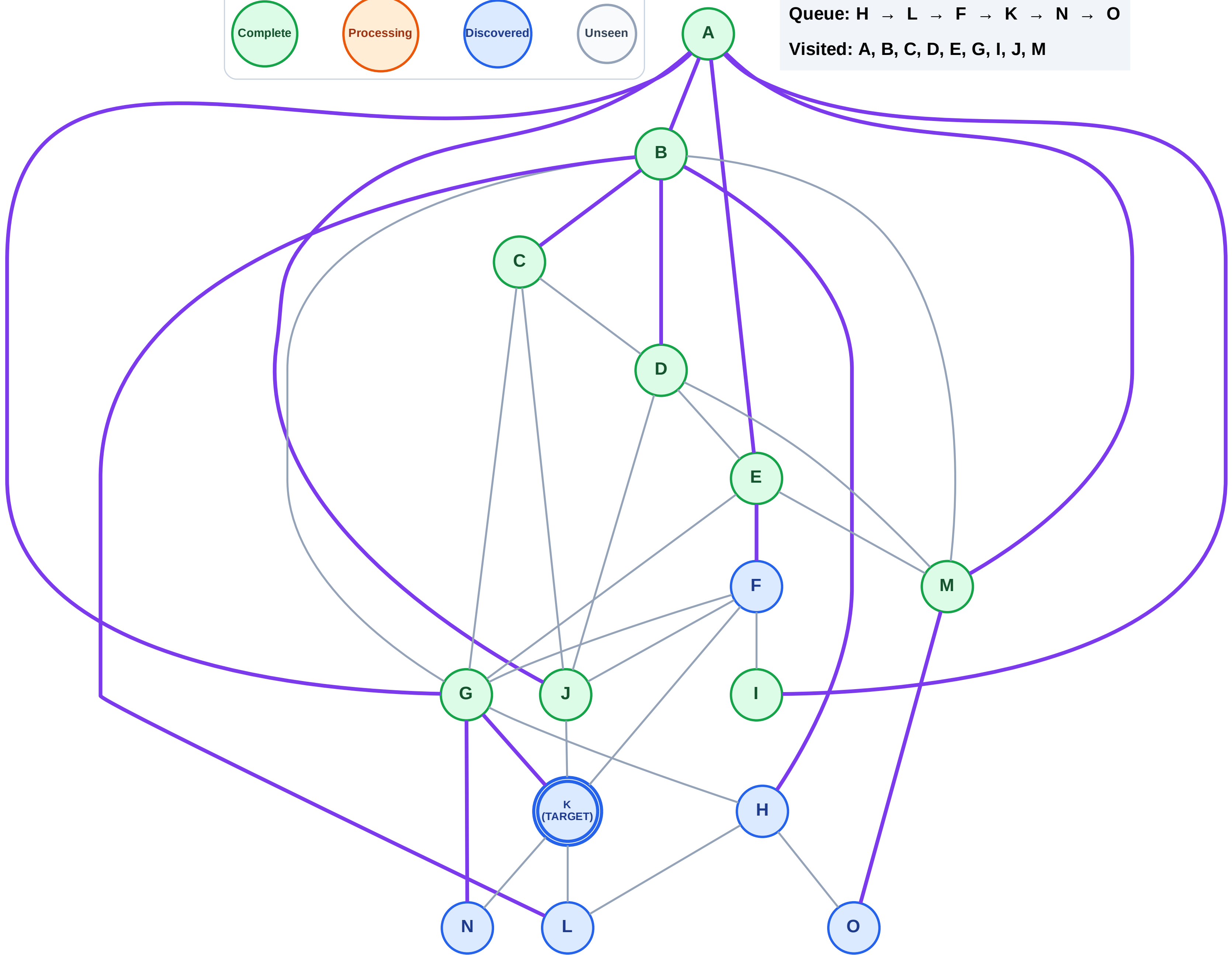
Step 19: No new neighbors from D

Legend



Queue: H → L → F → K → N → O

Visited: A, B, C, D, E, G, I, J, M



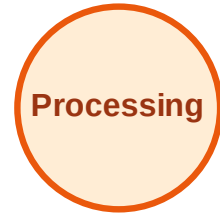
BFS Traversal

Step 20: Process node H

Legend



Complete



Processing



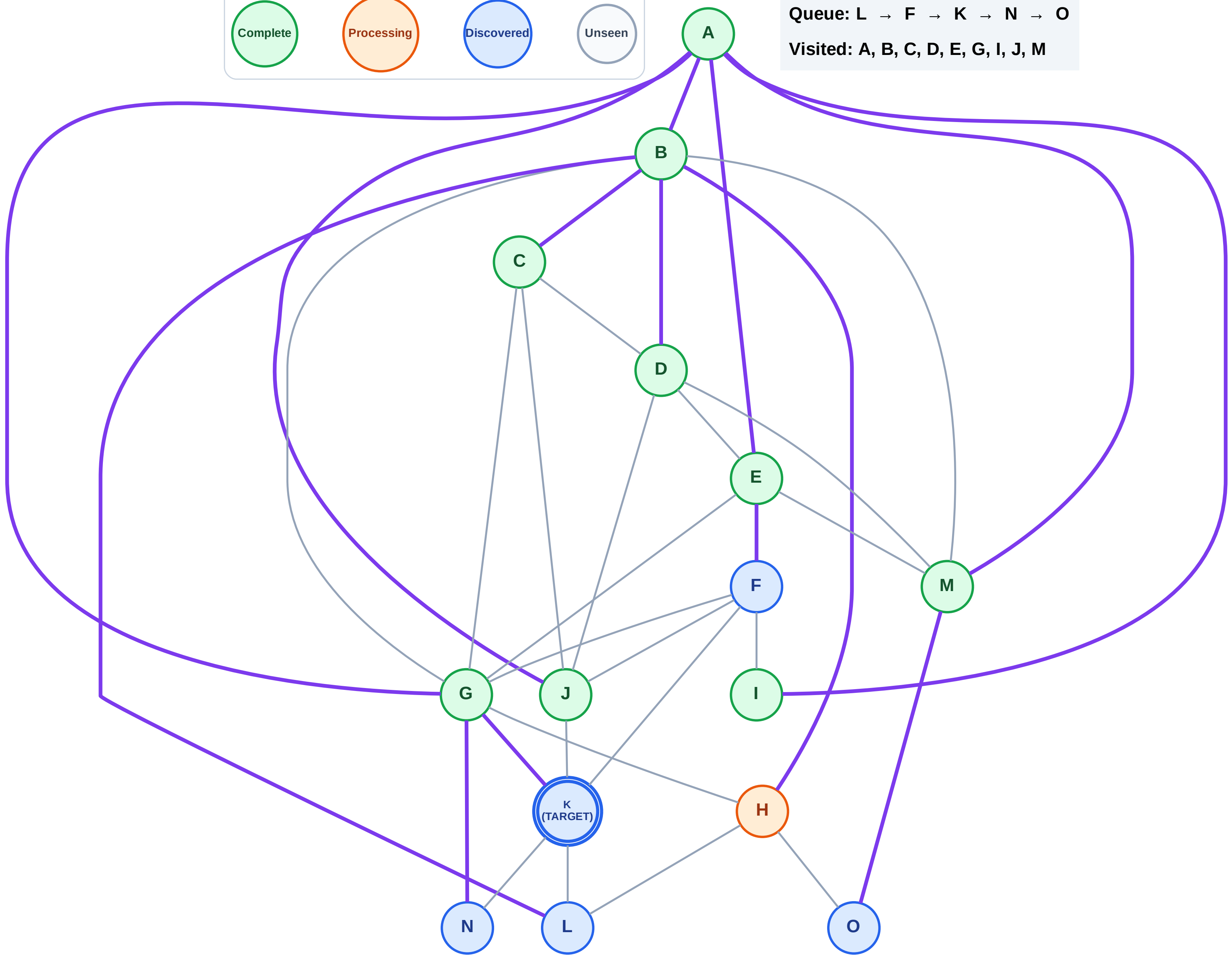
Discovered



Unseen

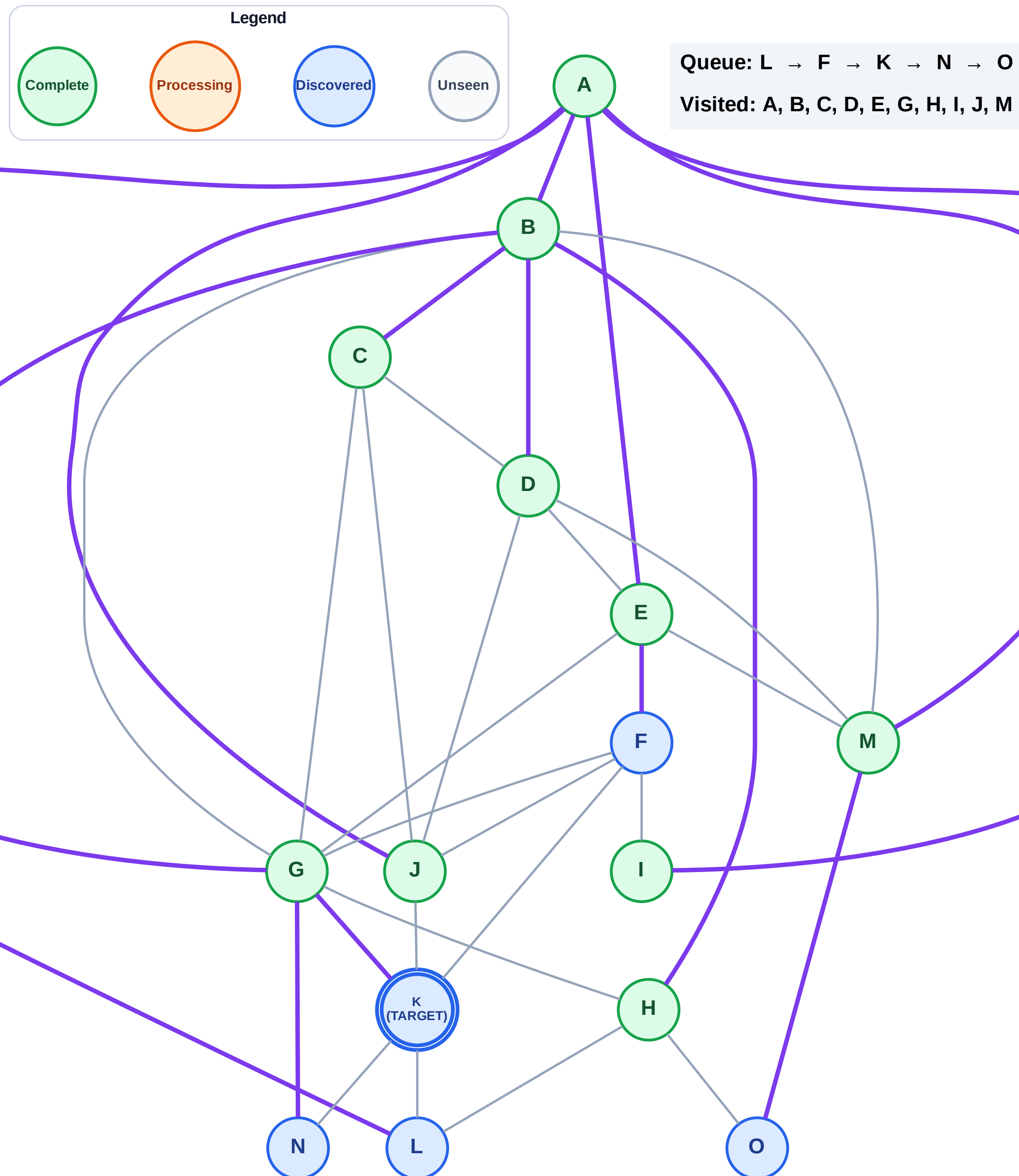
Queue: L → F → K → N → O

Visited: A, B, C, D, E, G, I, J, M



Step 21: No new neighbors from H

Step 21: No new neighbors from H



BFS Traversal

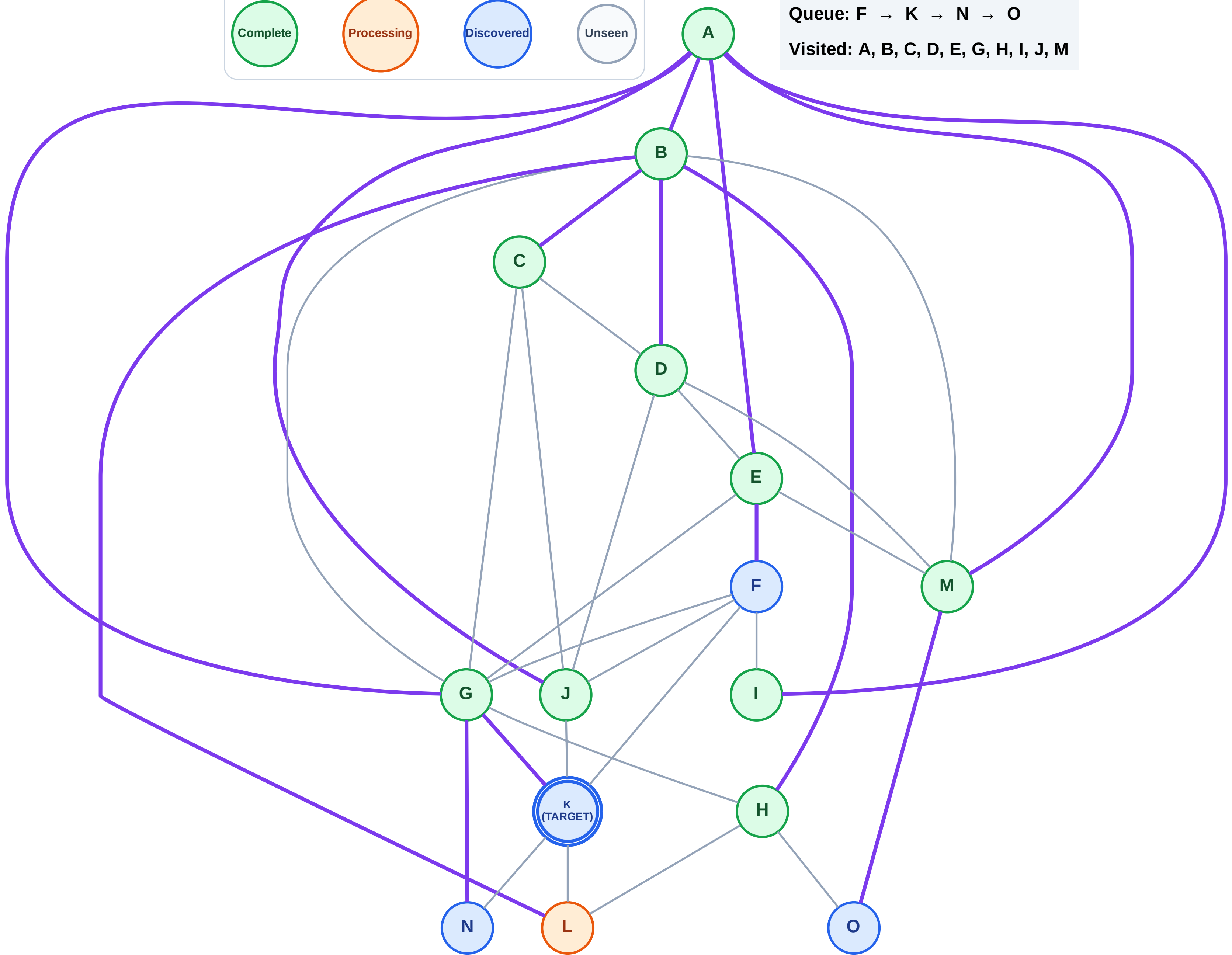
Step 22: Process node L

Legend



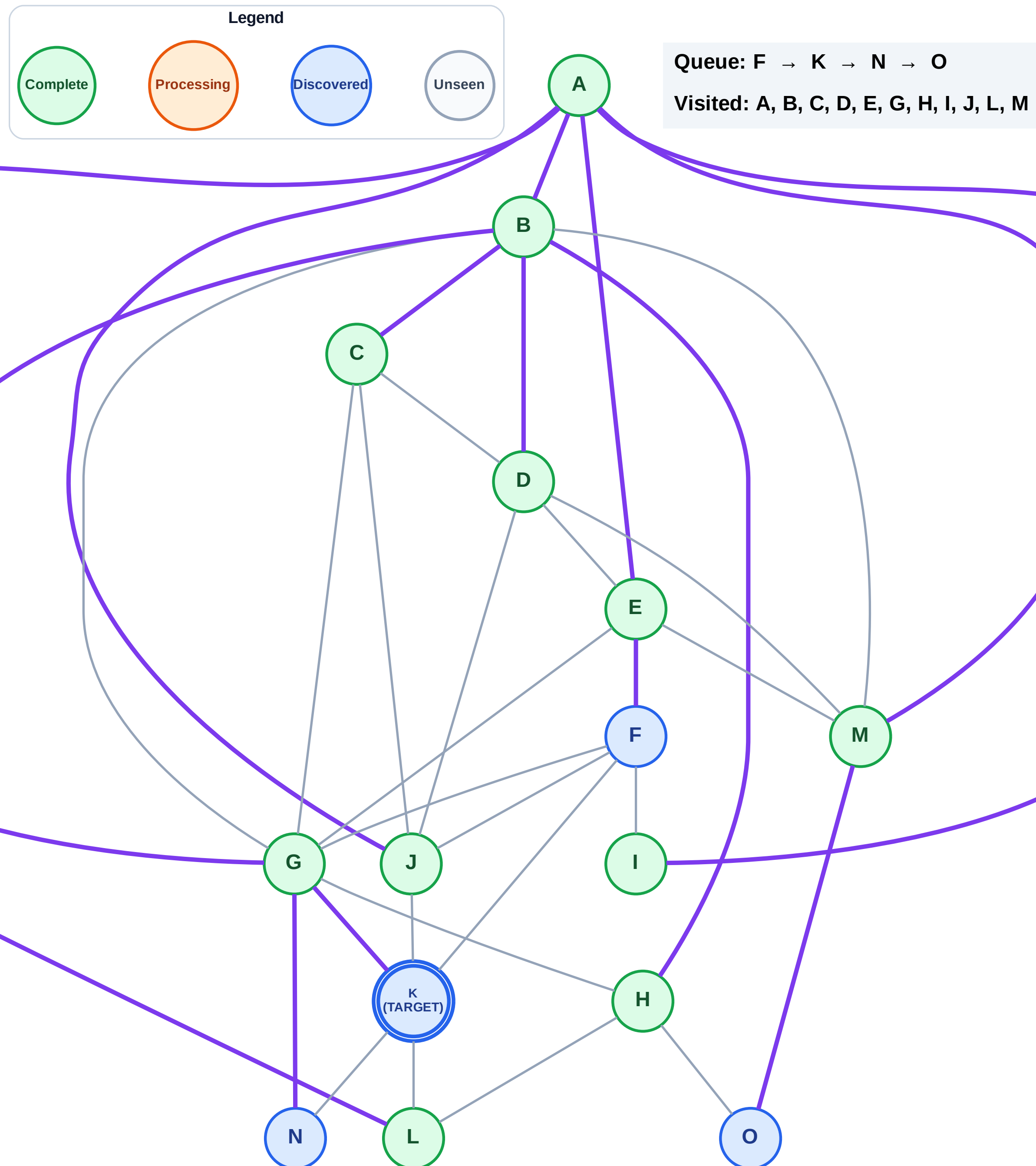
Queue: F → K → N → O

Visited: A, B, C, D, E, G, H, I, J, M



Step 23: No new neighbors from L

Step 23: No new neighbors from L



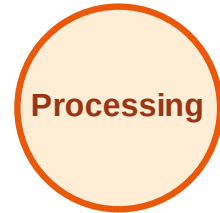
BFS Traversal

Step 24: Process node F

Legend



Complete



Processing



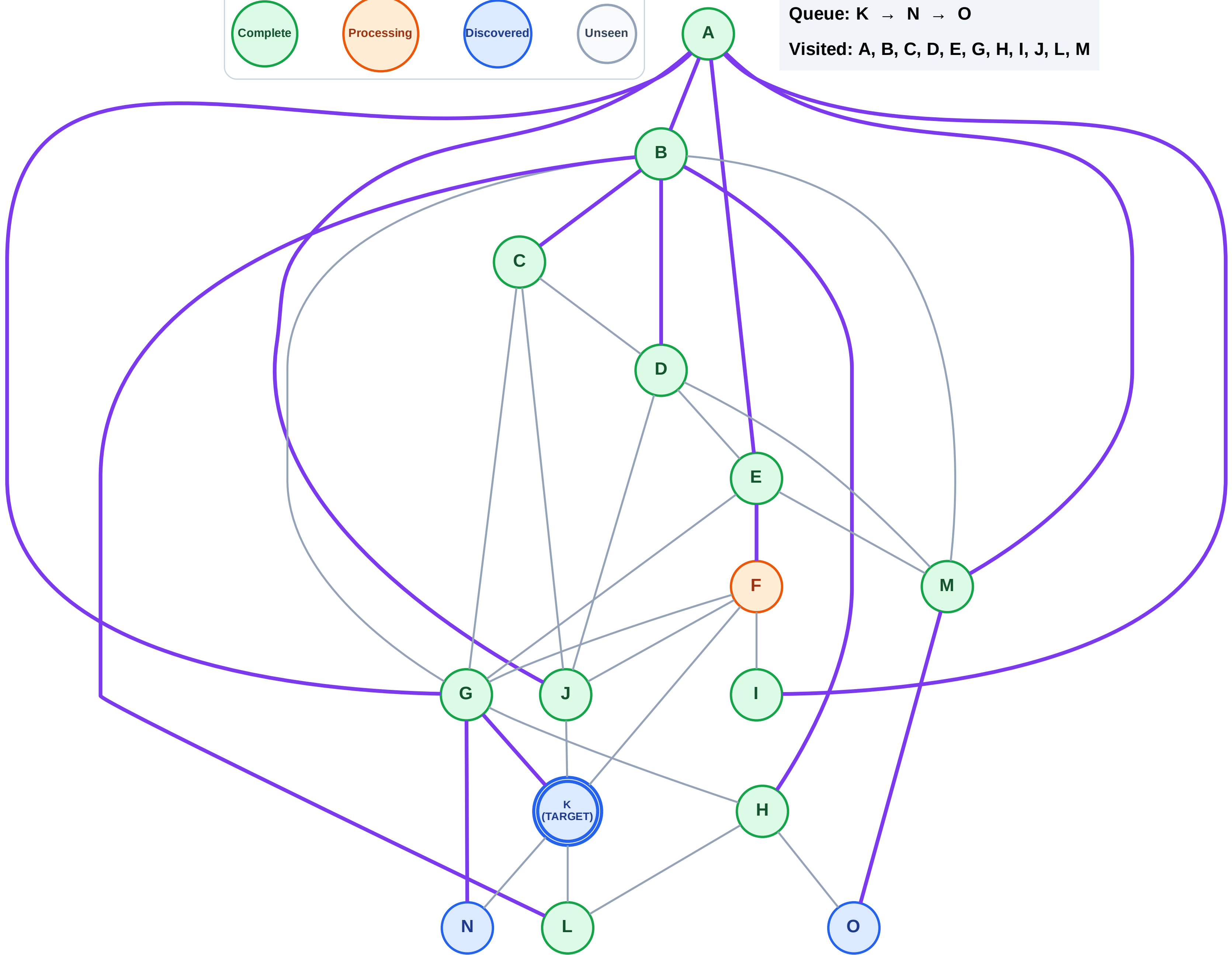
Discovered



Unseen

Queue: K → N → O

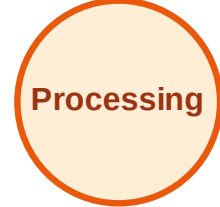
Visited: A, B, C, D, E, G, H, I, J, L, M



BFS Traversal

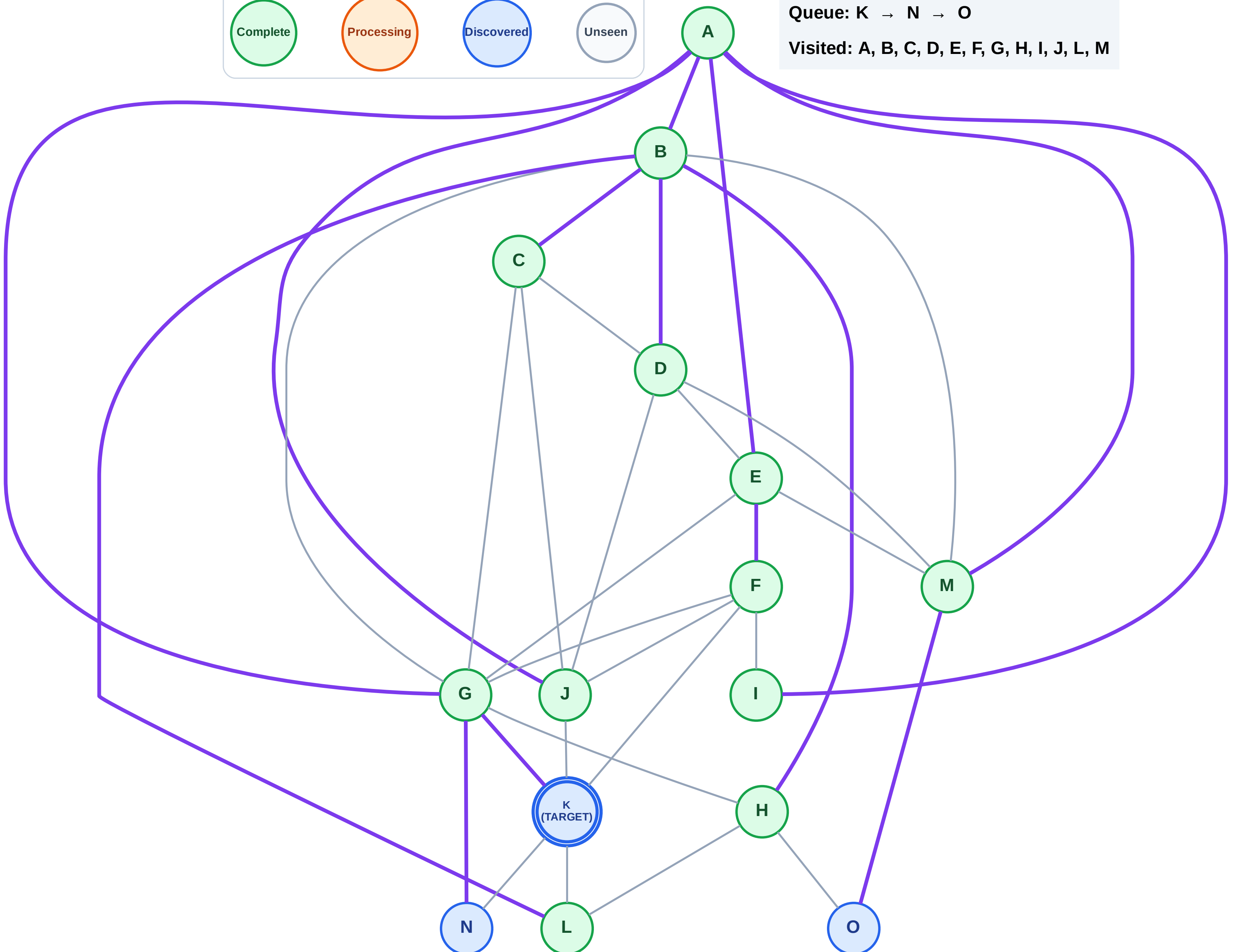
Step 25: No new neighbors from F

Legend



Queue: K → N → O

Visited: A, B, C, D, E, F, G, H, I, J, L, M



BFS Traversal

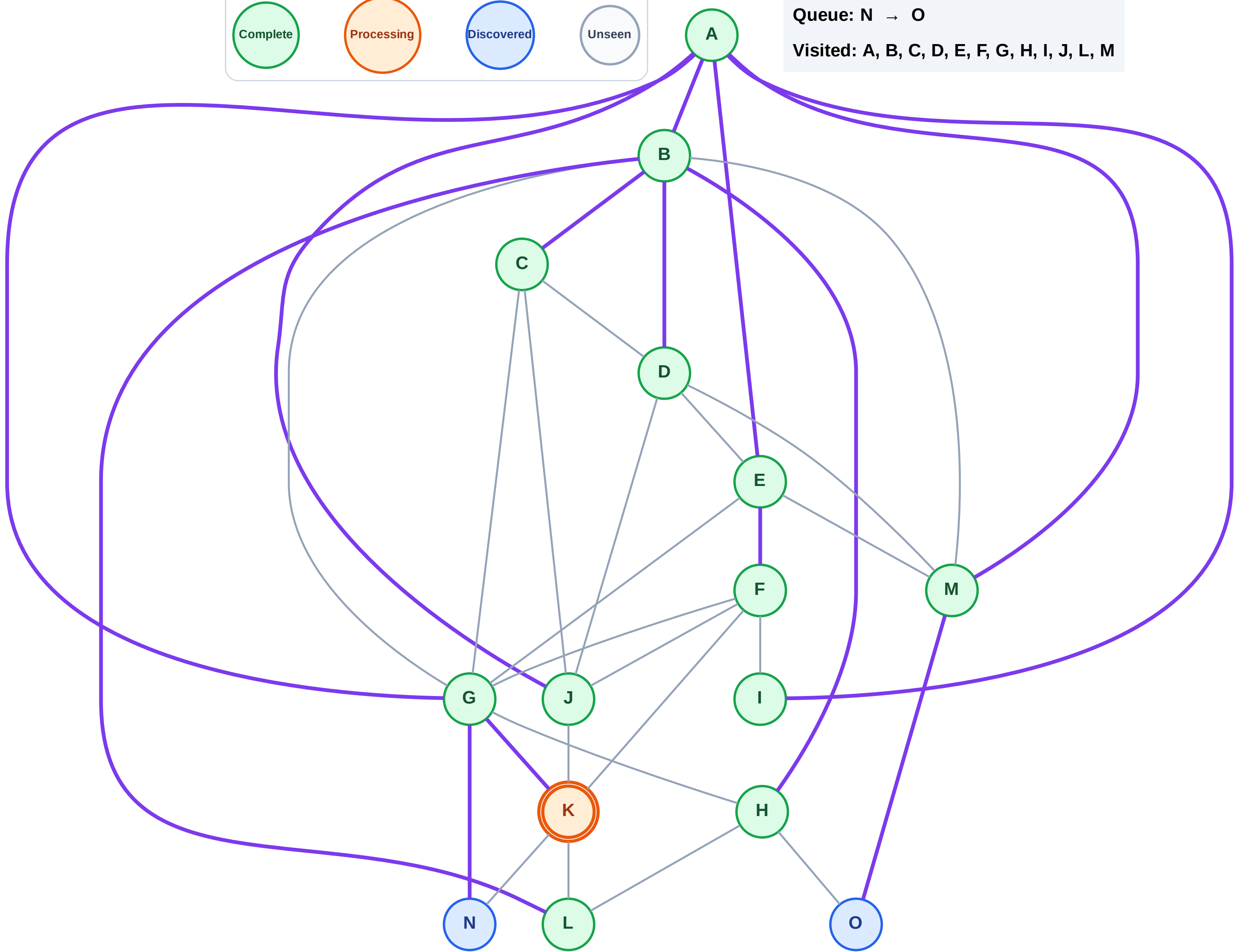
Step 26: Process node K

Legend



Queue: N → O

Visited: A, B, C, D, E, F, G, H, I, J, L, M



BFS Traversal

Step 27: TARGET NODE K REACHED! Path (2 hops): A → G → K

Legend

Complete

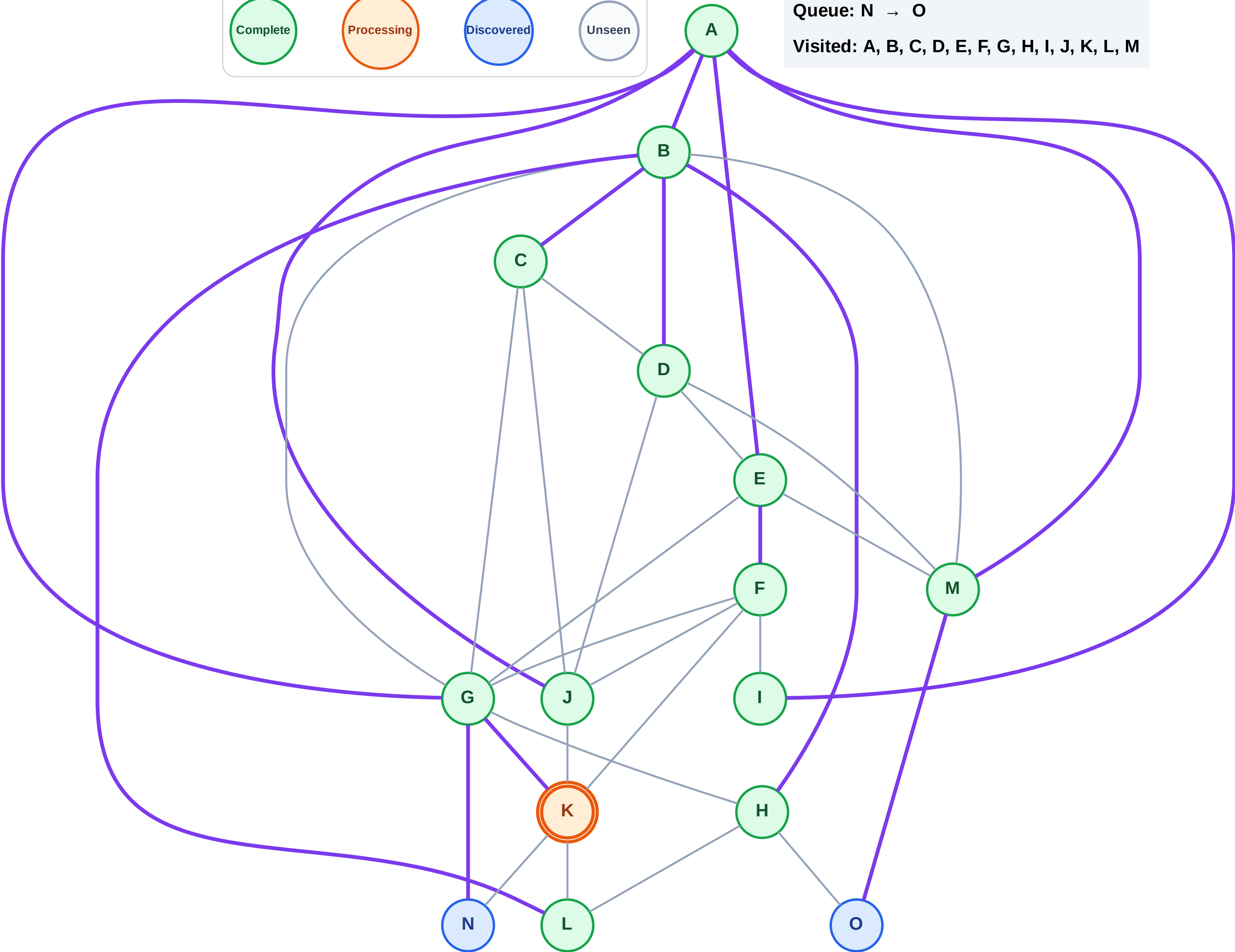
Processing

Discovered

Unseen

Queue: N → O

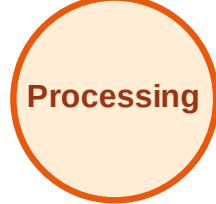
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M



DFS Traversal

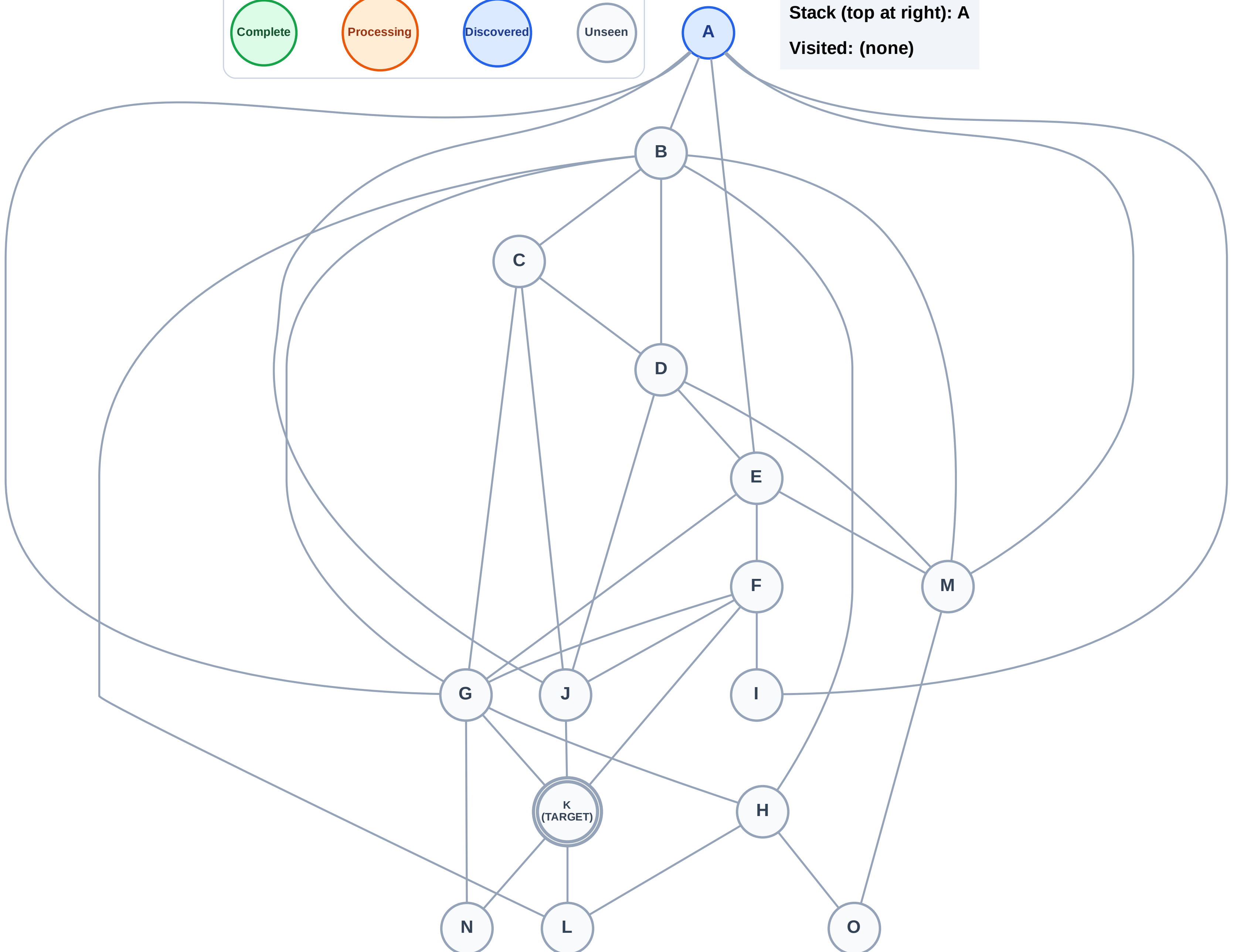
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

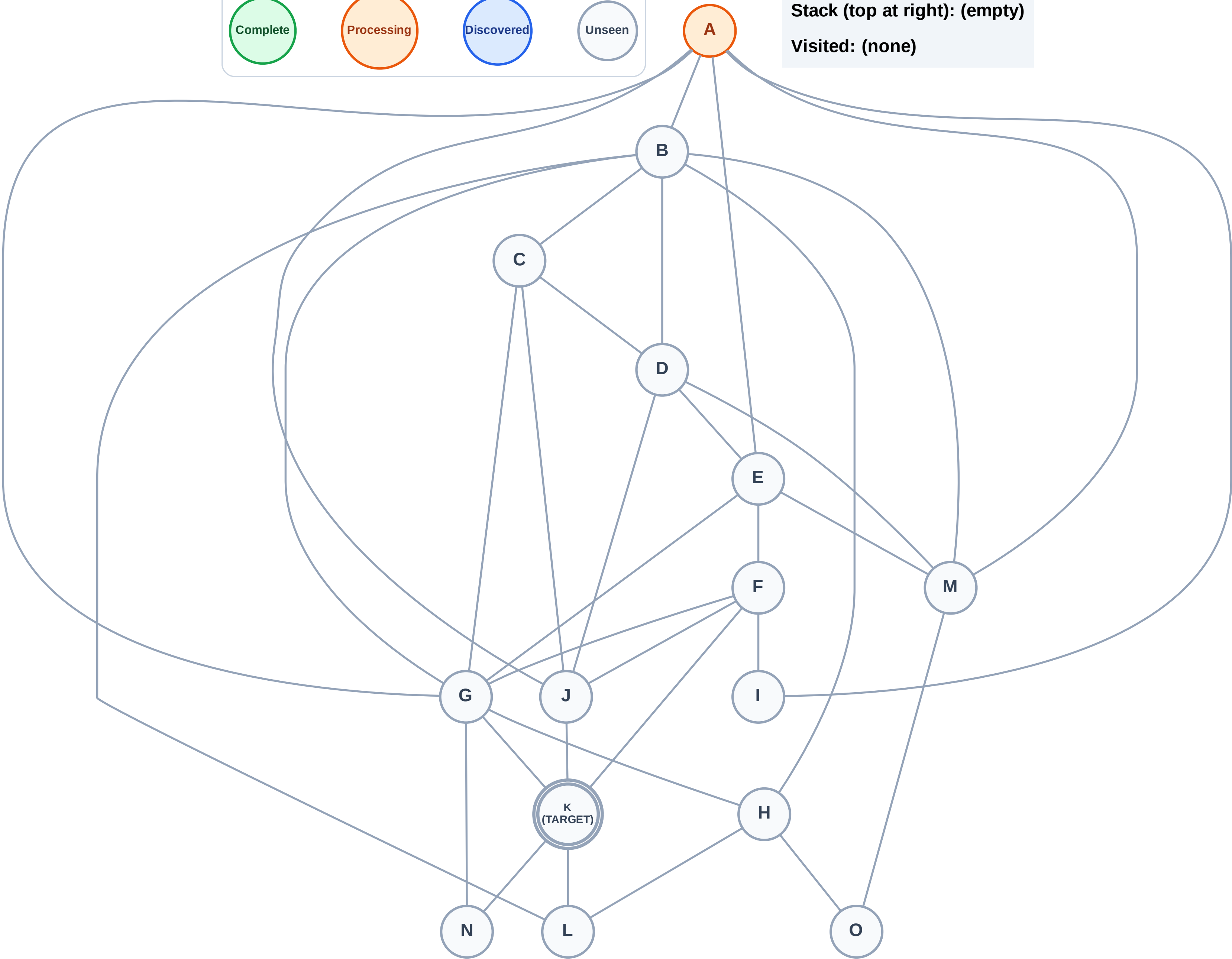
Complete

Processing

Discovered

Unseen

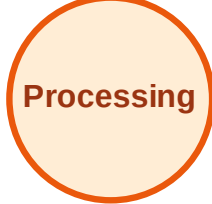
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

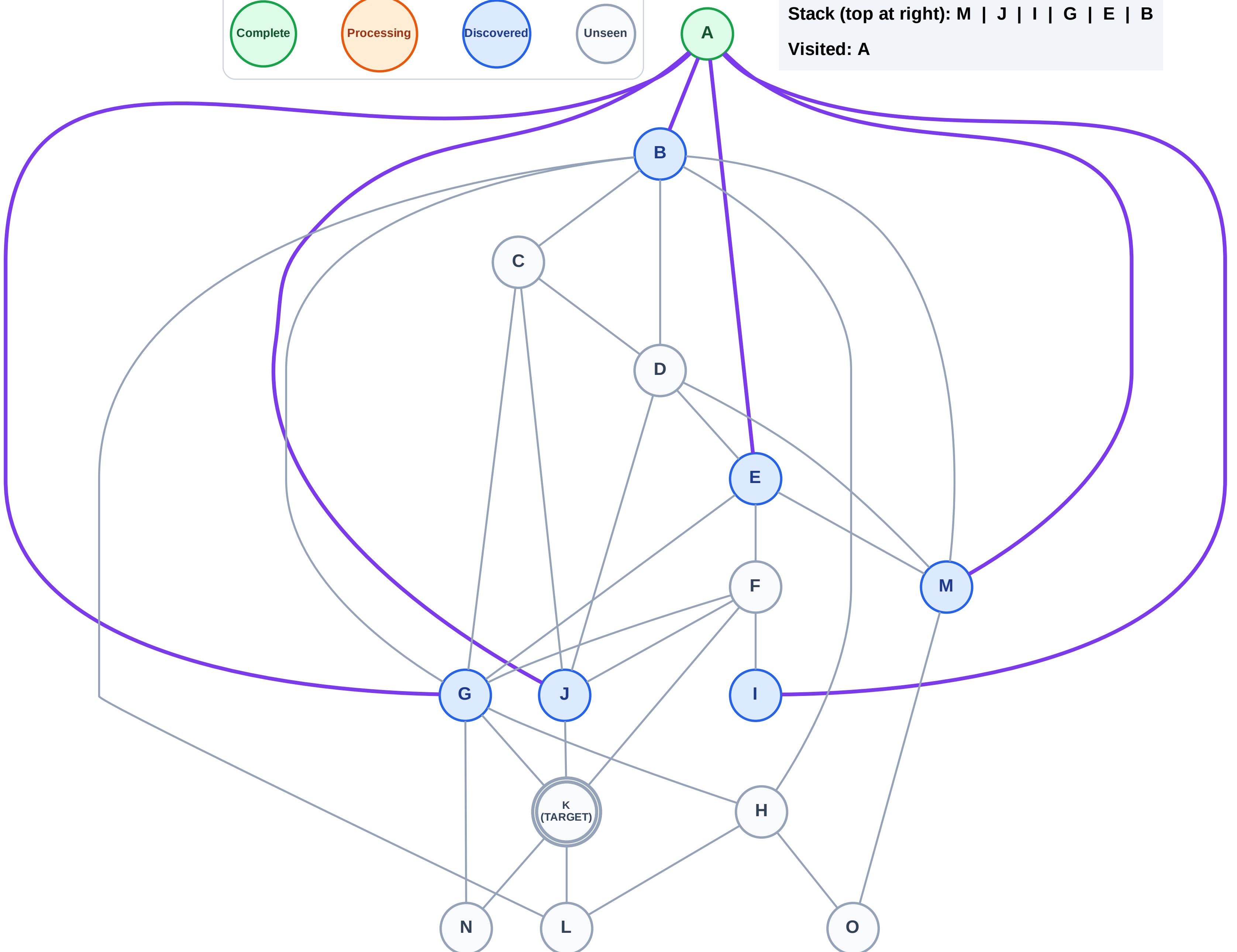
Step 3: Discover M, J, I, G, E, B from A

Legend



Stack (top at right): M | J | I | G | E | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

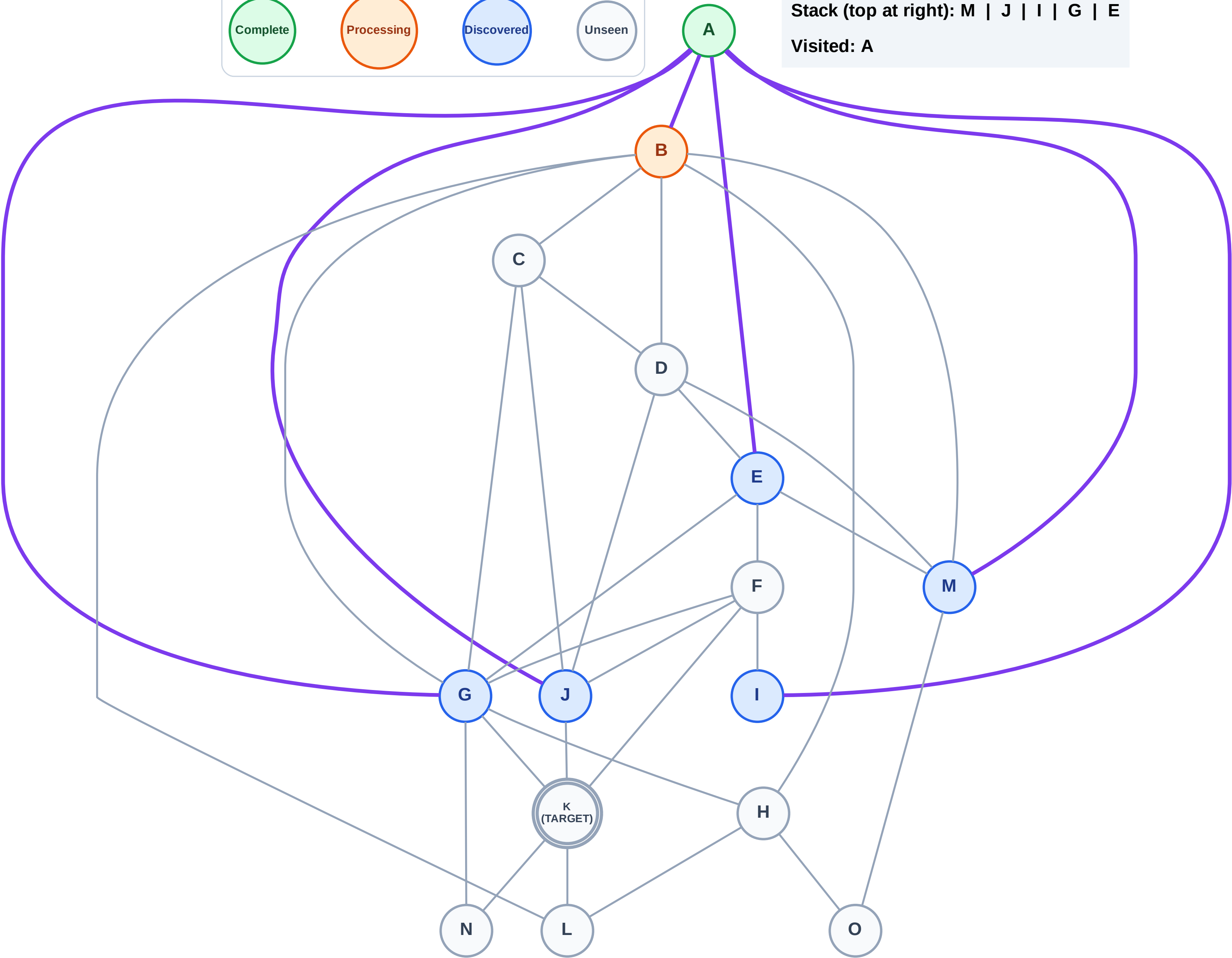
Processing

Discovered

Unseen

Stack (top at right): M | J | I | G | E

Visited: A



DFS Traversal

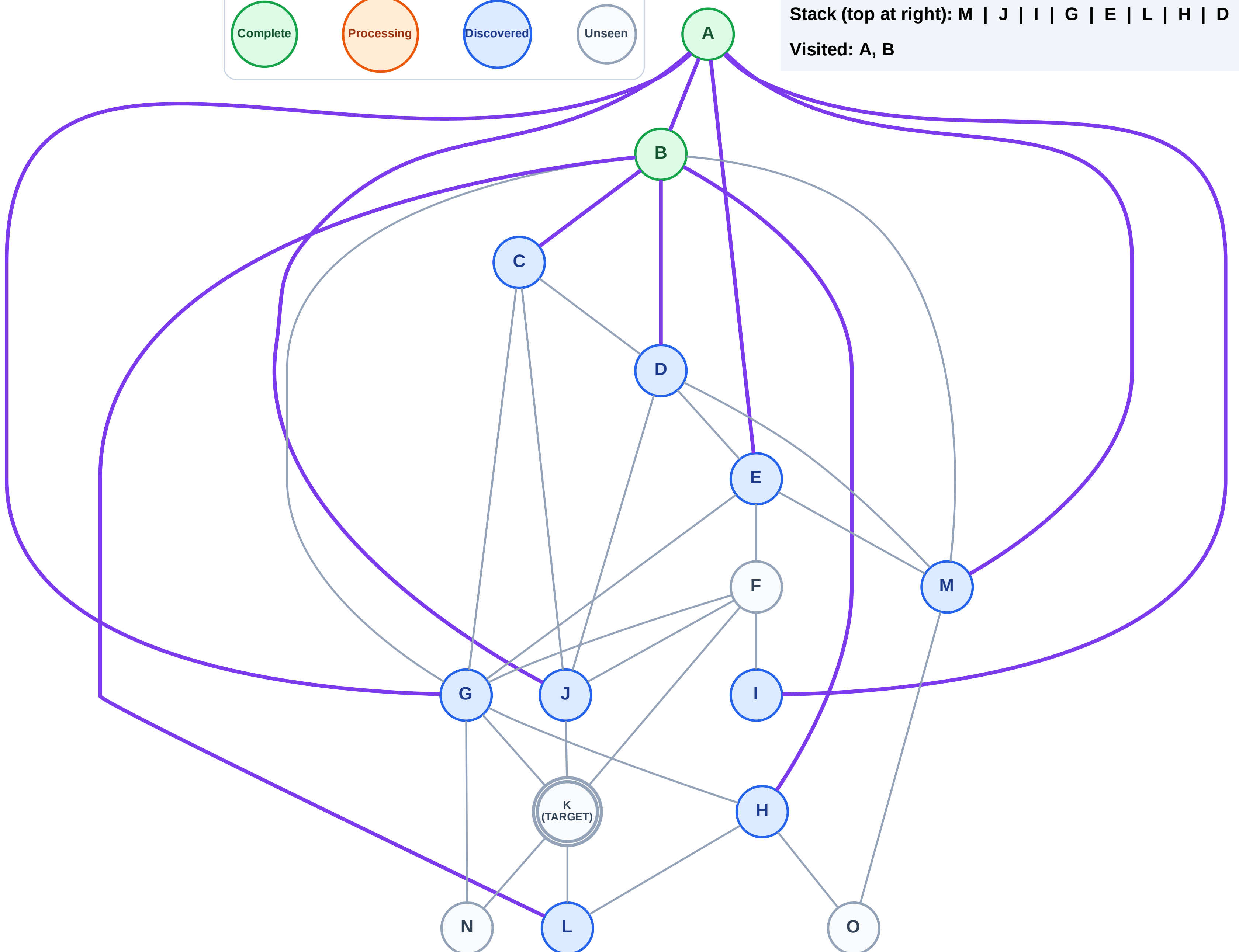
Step 5: Discover L, H, D, C from B

Legend



Stack (top at right): M | J | I | G | E | L | H | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

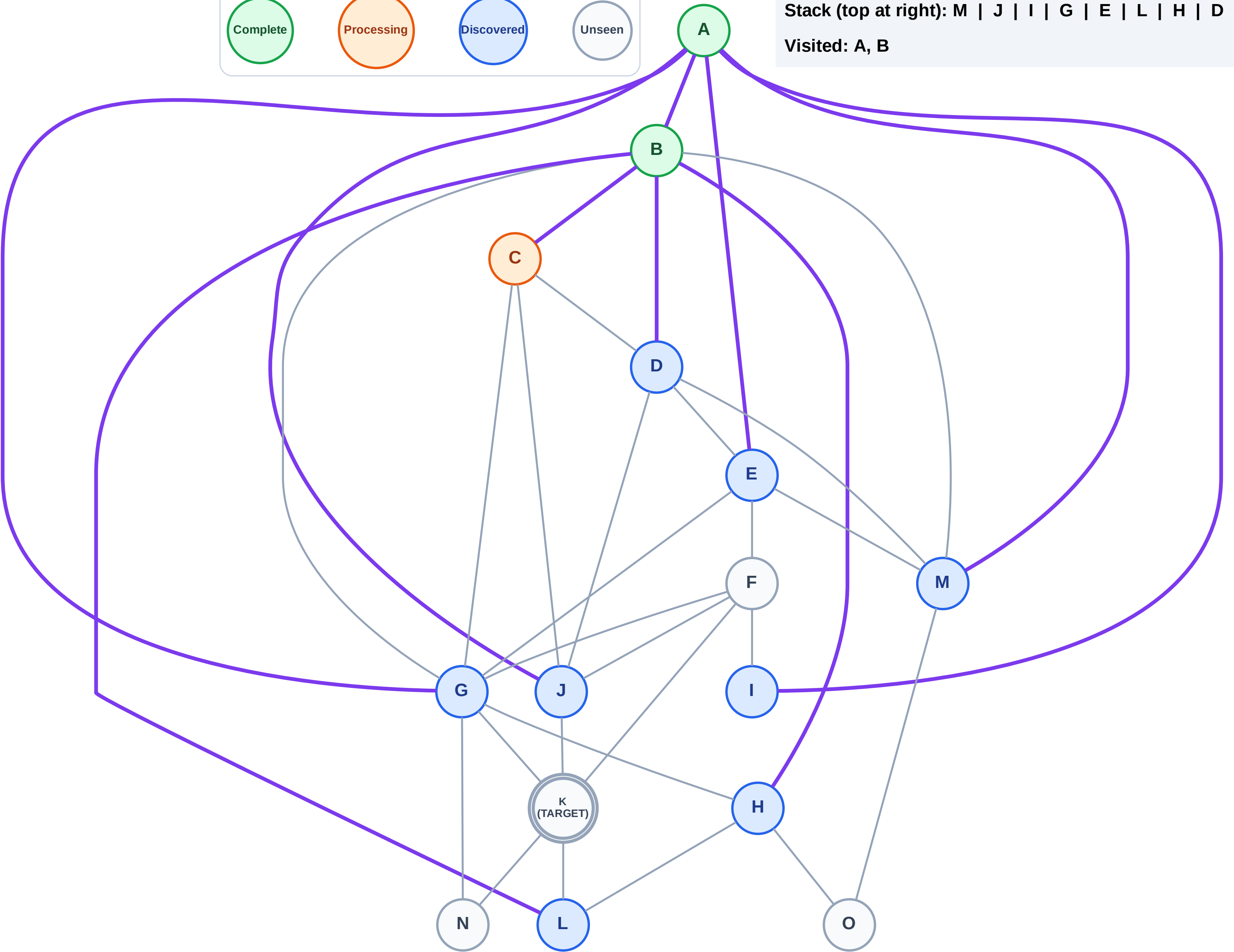
Processing

Discovered

Unseen

Stack (top at right): M | J | I | G | E | L | H | D

Visited: A, B



DFS Traversal

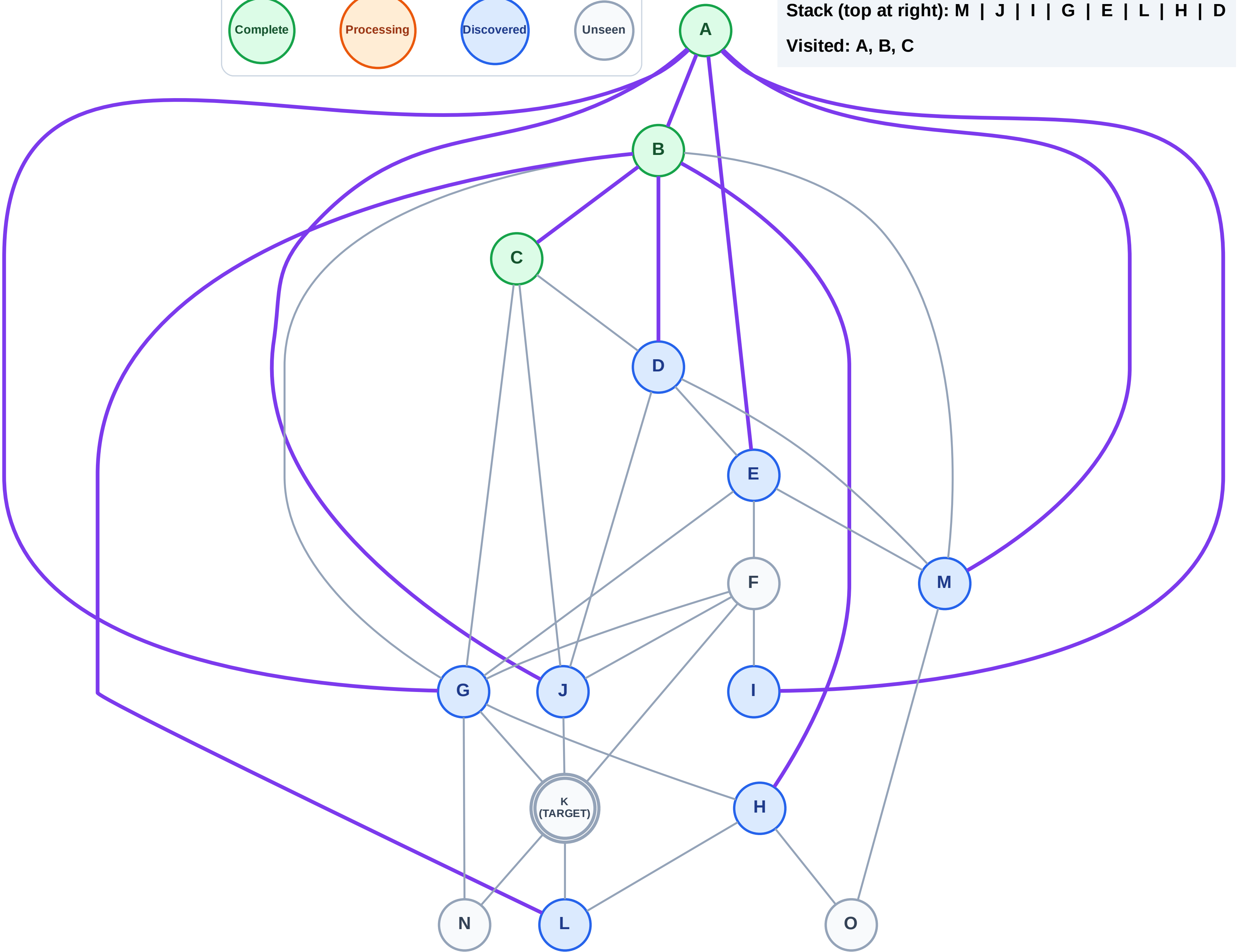
Step 7: No new neighbors from C

Legend



Stack (top at right): M | J | I | G | E | L | H | D

Visited: A, B, C



DFS Traversal

Step 8: Process node D

Legend

Complete

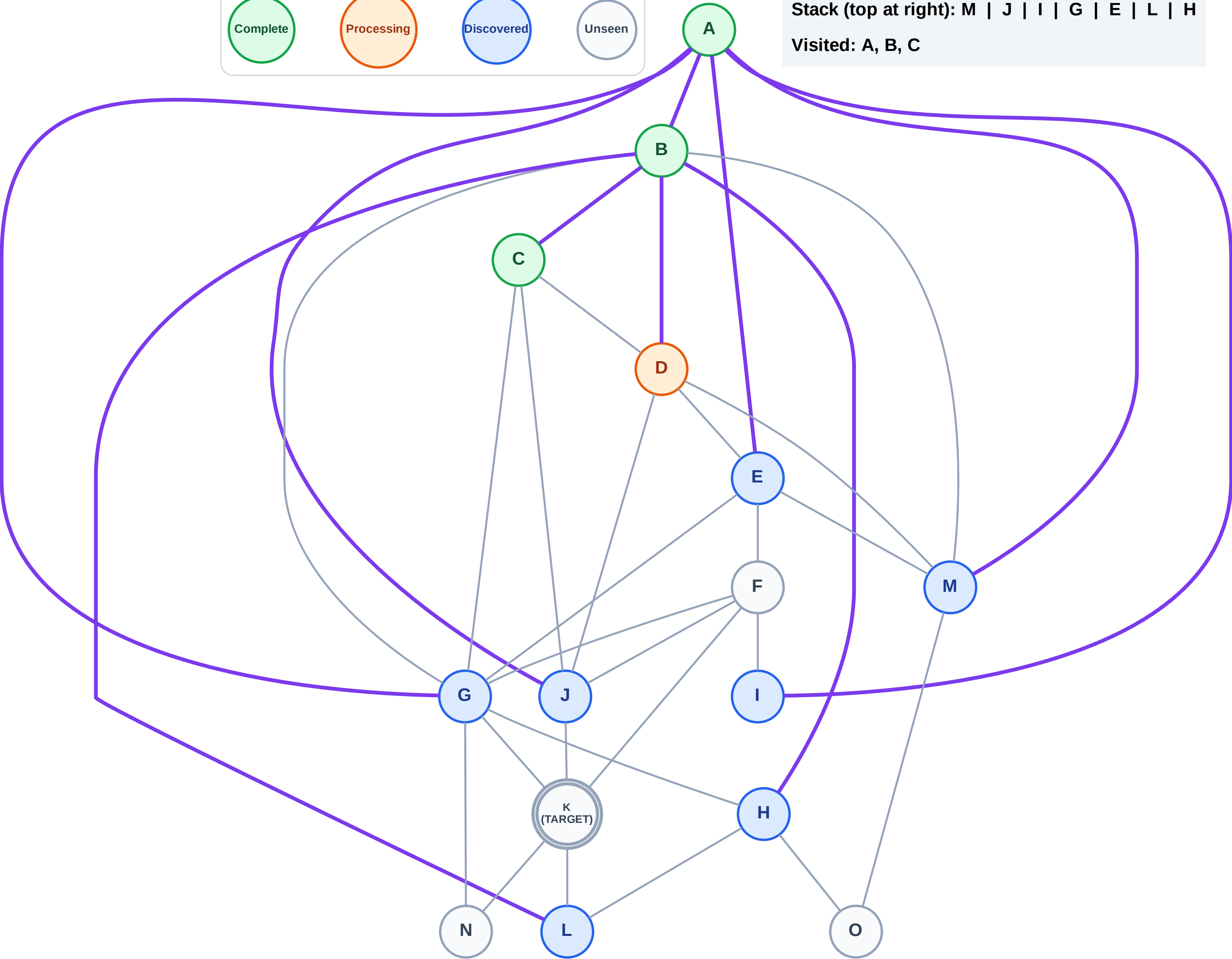
Processing

Discovered

Unseen

Stack (top at right): M | J | I | G | E | L | H

Visited: A, B, C

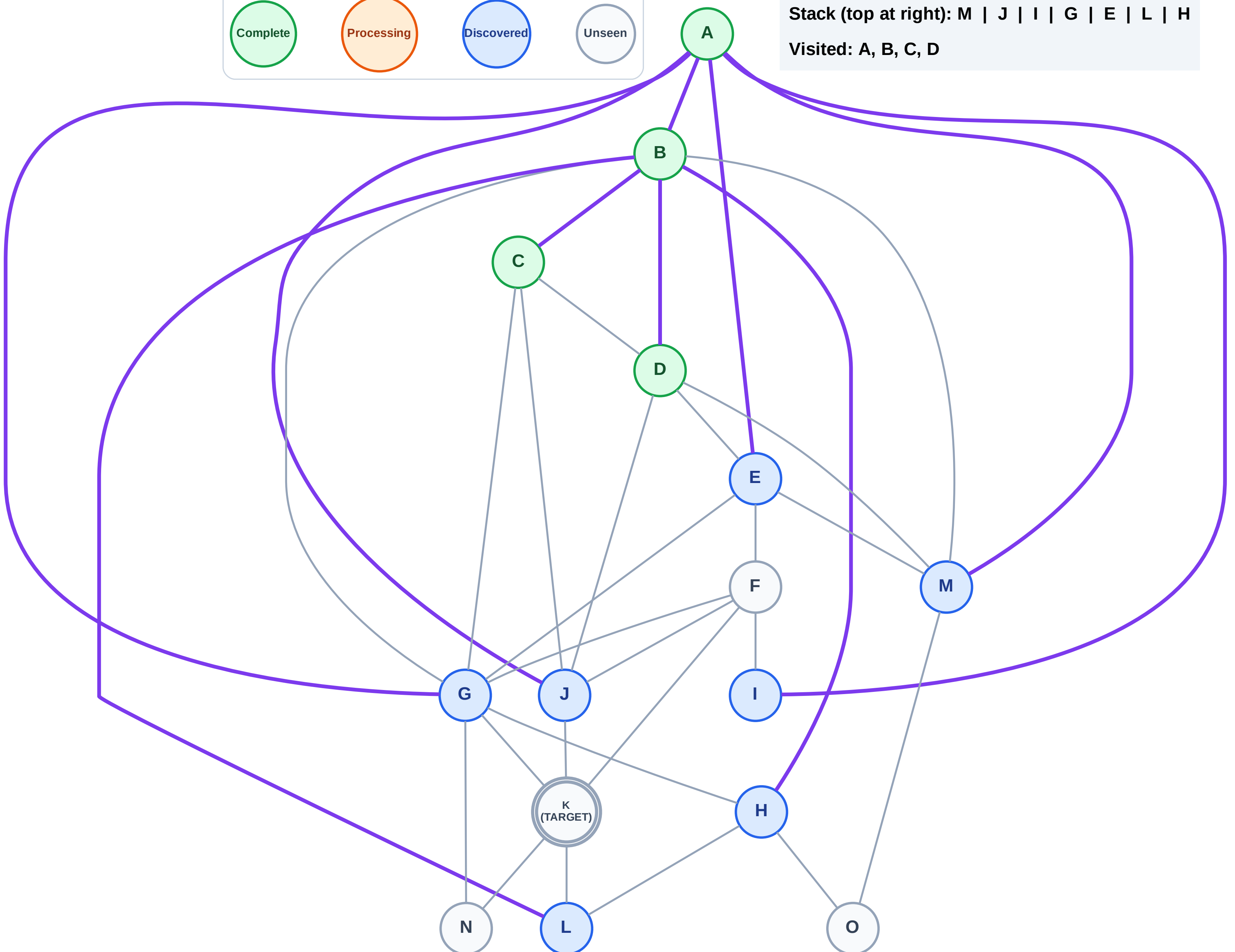


Step 9: No new neighbors from D

Legend

Unseen

Visited: A, B, C, D



DFS Traversal

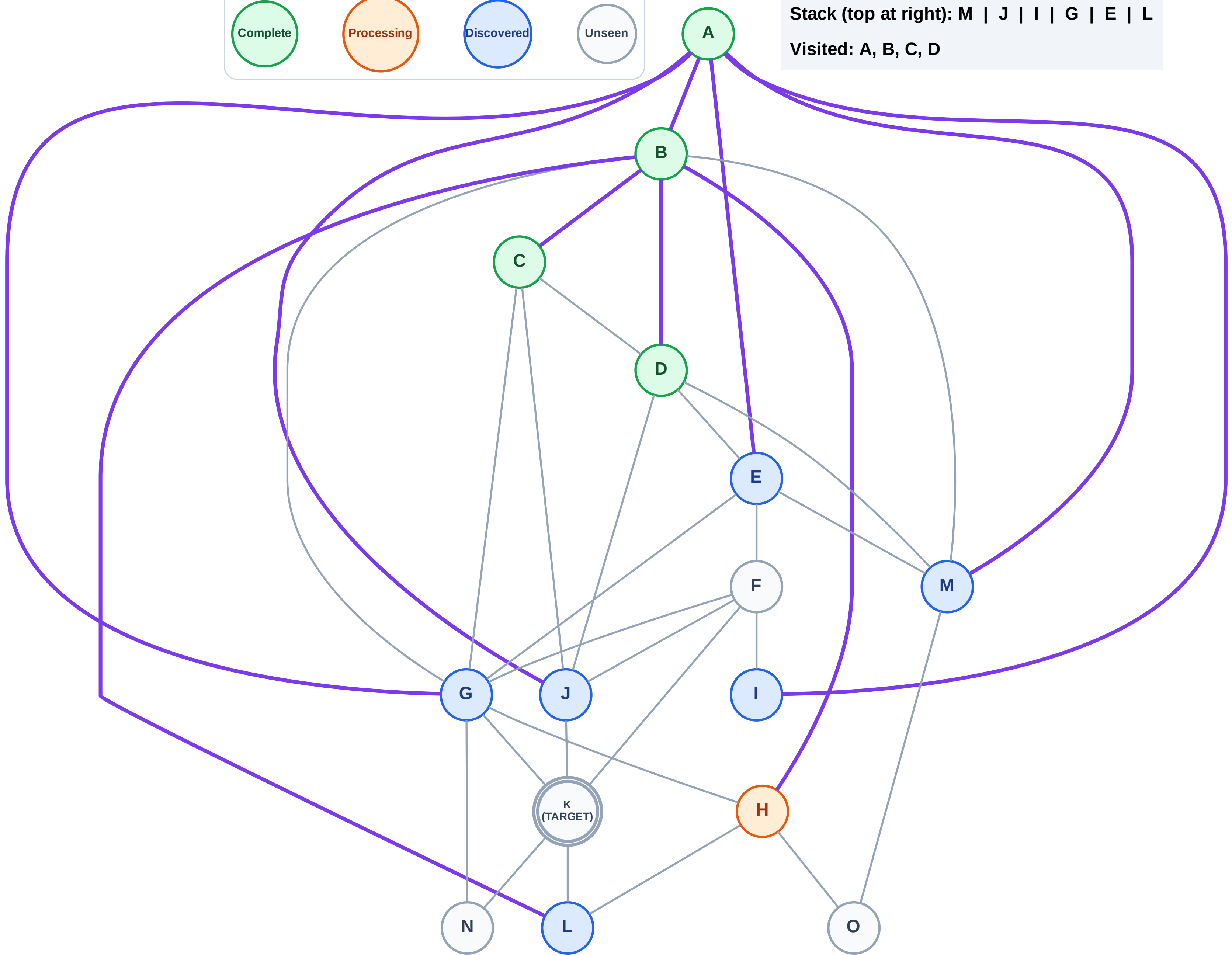
Step 10: Process node H

Legend



Stack (top at right): M | J | I | G | E | L

Visited: A, B, C, D



DFS Traversal

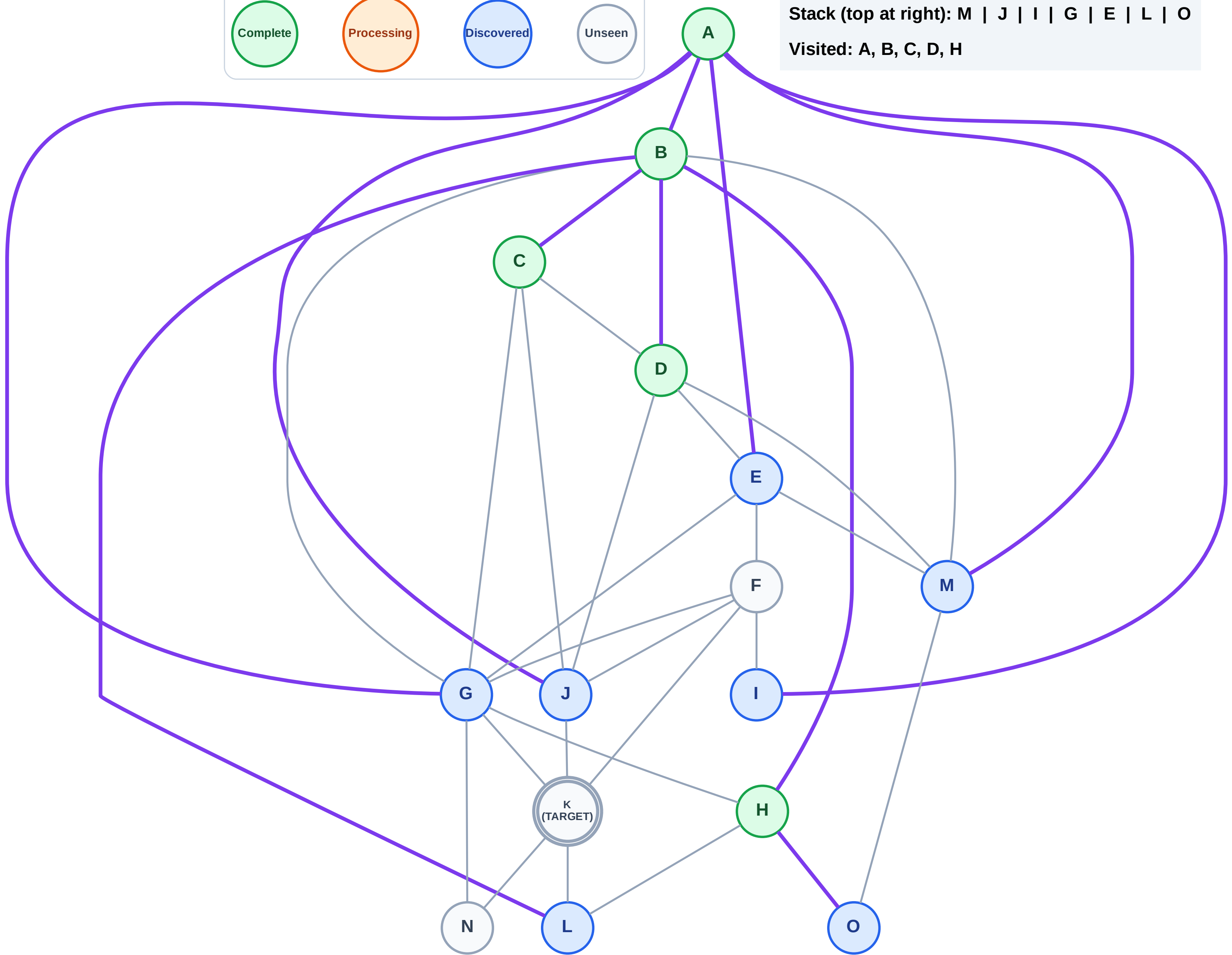
Step 11: Discover O from H

Legend



Stack (top at right): M | J | I | G | E | L | O

Visited: A, B, C, D, H



DFS Traversal

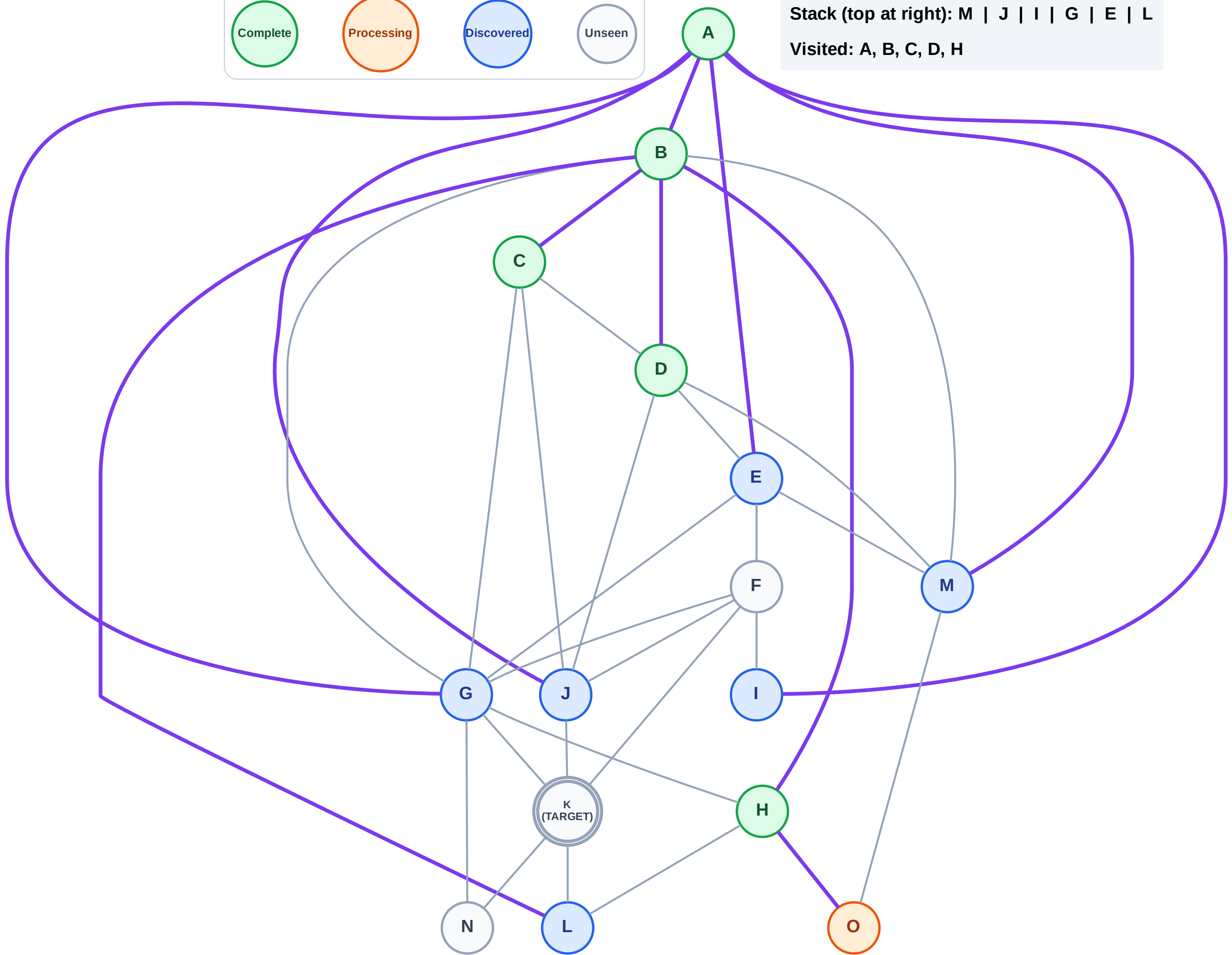
Step 12: Process node O

Legend



Stack (top at right): M | J | I | G | E | L

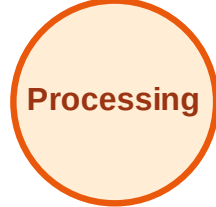
Visited: A, B, C, D, H



DFS Traversal

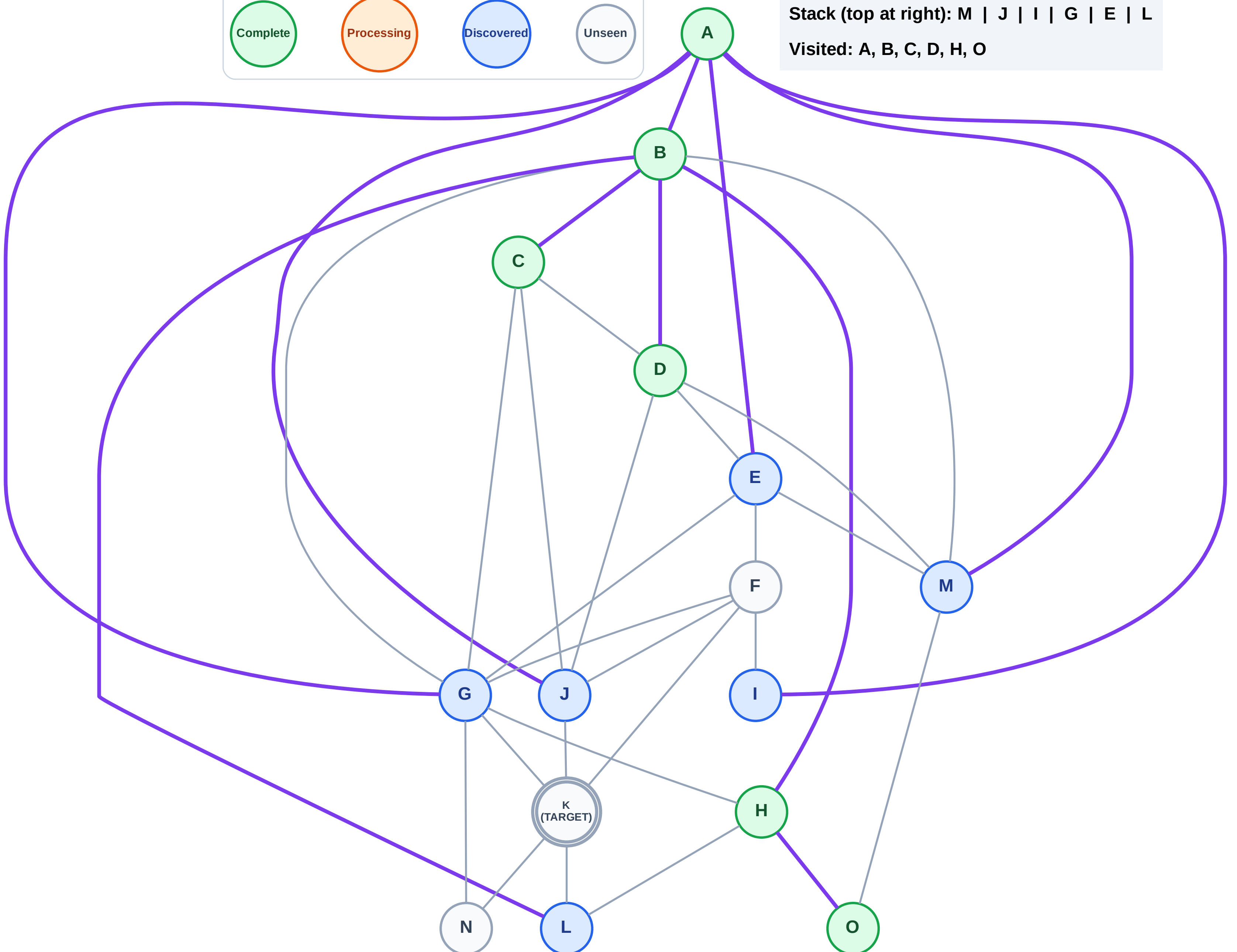
Step 13: No new neighbors from O

Legend



Stack (top at right): M | J | I | G | E | L

Visited: A, B, C, D, H, O



DFS Traversal

Step 14: Process node L

Legend

Complete

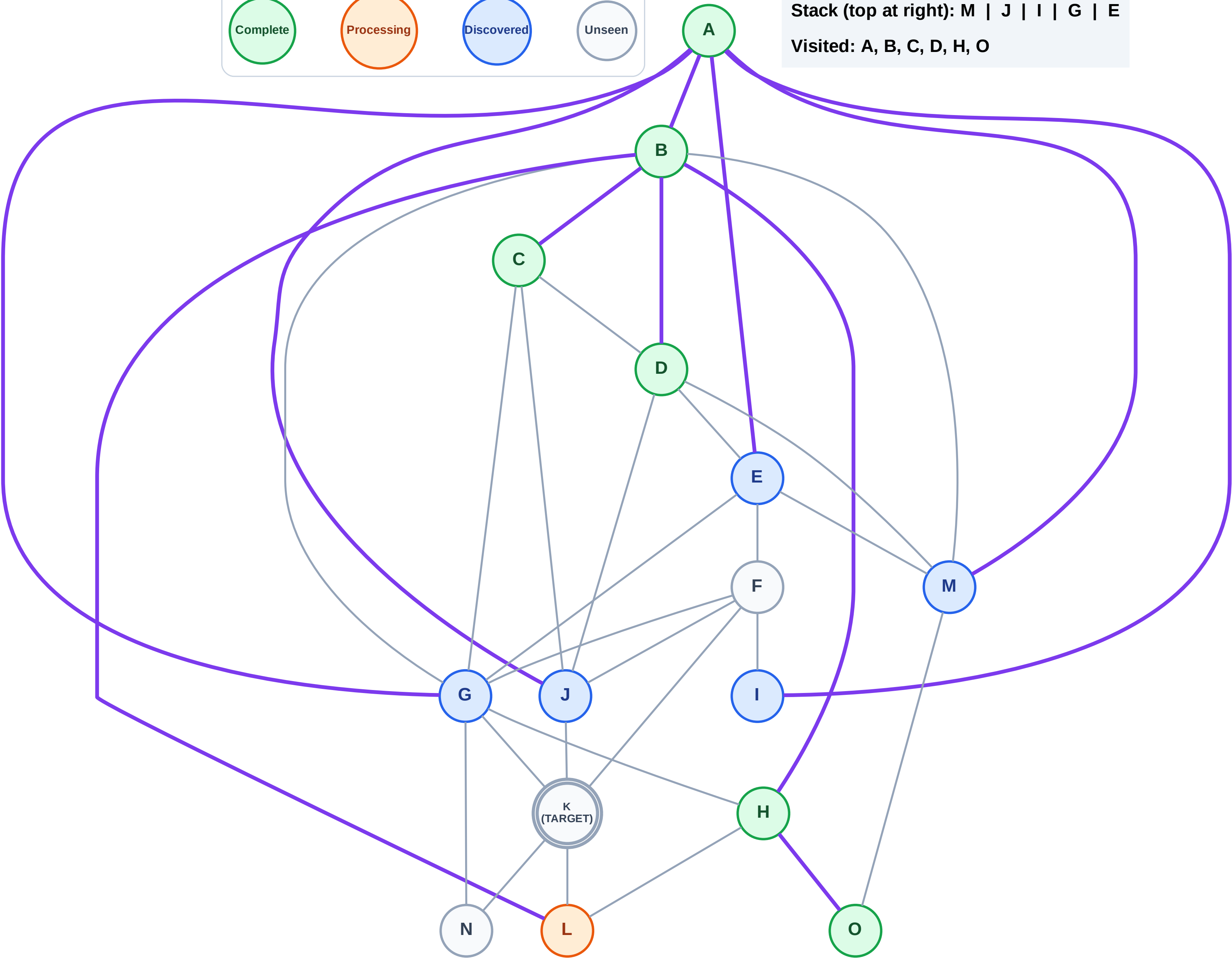
Processing

Discovered

Unseen

Stack (top at right): M | J | I | G | E

Visited: A, B, C, D, H, O



DFS Traversal

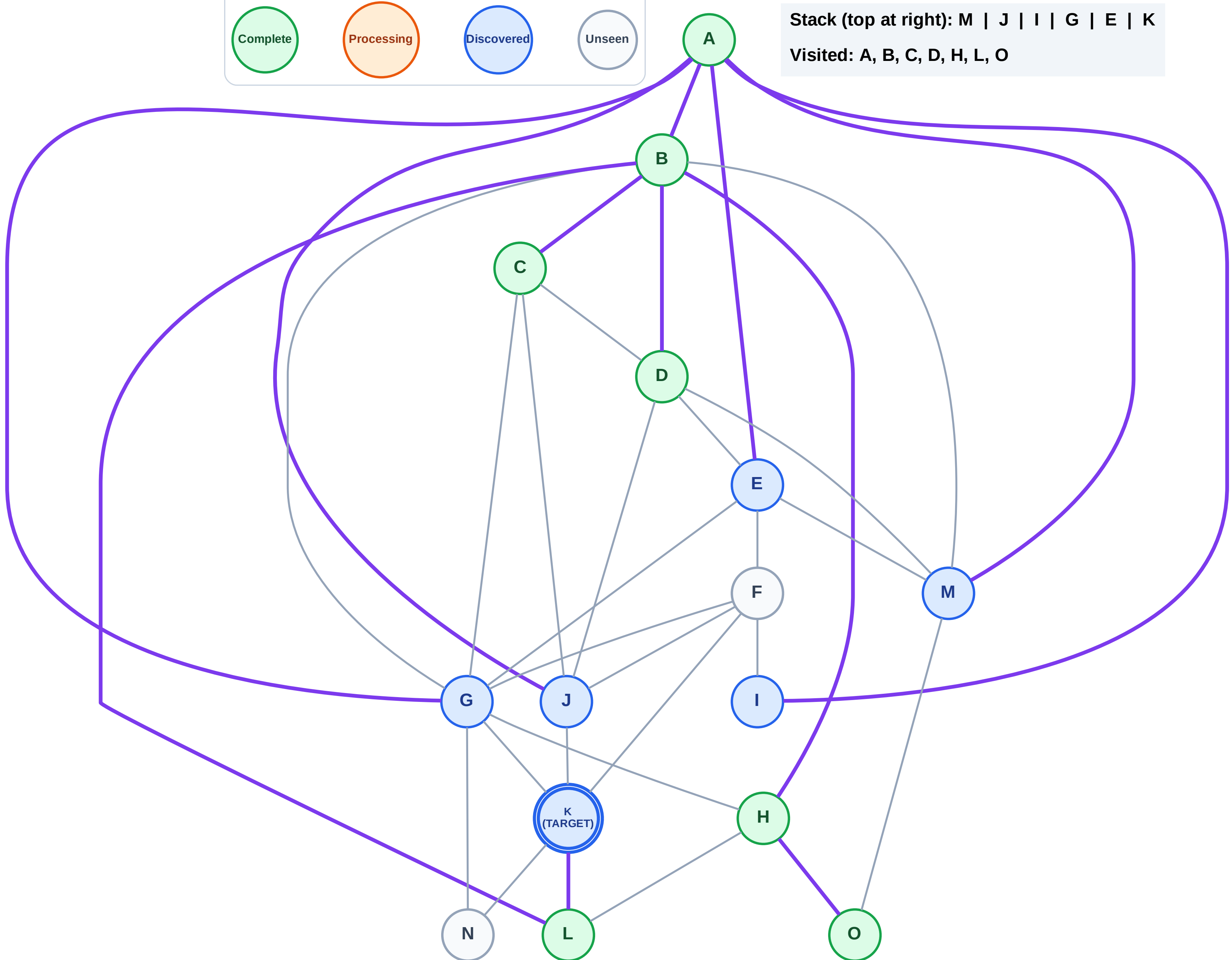
Step 15: Discover K from L

Legend



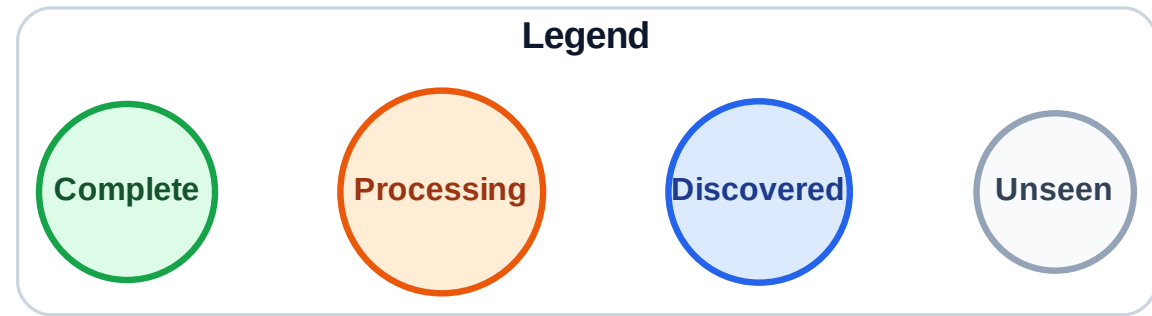
Stack (top at right): M | J | I | G | E | K

Visited: A, B, C, D, H, L, O



Step 16: Process node K

Page 10 of 10



DFS Traversal

Step 17: TARGET NODE K REACHED! Path (3 hops): A → B → L → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | J | I | G | E
Visited: A, B, C, D, H, K, L, O

